Priced Runtime Control over Frozen Agent Harnesses: a Learned Orchestrator, an Off-Policy Promotion Gate, and Drift Metrics

Technical report, draft v0.1

Navid Mehrdad

October 2026

Abstract

Agent harnesses contain per-request decisions that nobody prices: which model tier answers, whether a sub-result is verified, whether an answer is refined, which specialist is delegated to. The field evolves harnesses at design time; this report studies the complementary runtime problem. We keep the harness frozen and learn a small, cost-priced controller at its decision points, with three pieces: a softmax-linear (or small-LLM) policy trained offline from logged episodes under a budget whose price λ is a learned Lagrange multiplier; an off-policy promotion gate that admits a candidate only when doubly-robust (DR) estimates with bootstrap confidence intervals certify quality and cost; and priced backward/forward-transfer metrics for the harness changing under the controller. Six experiments are reported, with negative results stated as results. On a simulator the gate promotes a candidate whose DR interval covers the ground truth, and a gated retrain recovers $+0.033$ of the -0.045 that drift costs a frozen controller. On CRMArena-Pro (150 tasks $\times$ 3 tiers, \$59.90) the per-task oracle beats every fixed tier at lower cost, and with three times the data the learned per-category policy beats every fixed tier held-out on quality and cost at once (0.623 at \$0.082 against the frontier tier's 0.616 at \$0.143), while the budgeted controller exceeds frontier quality at 27% less cost; at 50 tasks the same policies memorised and lost, and the gate said so. On the realised one-arm log the doubly-robust gate promotes on one draw and reverts on another, 10% of redraws promote, and certifying a \$0.005 budget margin remains a thousand-task matter. A GRPO-trained 1.7B-parameter controller never found the budgeted optimum from near-identical rollout groups; an SFT warm start did the work and GRPO's contribution was the price ($\lambda \approx 0.44$). A planner/worker/verifier decomposition scores below a single mid-tier agent (0.480 vs 0.608), its verifier flags 92% of wrong and 96% of correct episodes, and an orchestration reward model on process features has out-of-fold AUROC 0.469 against 0.821 for the task category alone. Priced self-refinement after a cheap answer is dominated by buying a better tier up front (-0.174 ± 0.085 paired at equal cost). The centrepiece is a continual orchestrator over a frozen multi-agent structure on a simulator calibrated to the real logs, under four pre-registered harness stages over 40 days: the online learner has the lowest regret of every learned arm (1.26 vs 2.60 frozen and 3.03–3.46 for static routers), recovers after every stage, and has bounded, mostly positive orchestrator-level BWT; but a fixed mode (`never_verify`, regret 0.56; 0.74 ± 0.08 against the learner's 1.06 ± 0.21 over 5 seeds) beats it over the whole stream. That headline is a cold-start artefact: the learner spends 12 days on the uniform exploration policy before its first promotion, crosses the fixed mode around day 22–25 and wins every stage window after that (held-out regret from day 25: 0.22 vs 0.50), and its post-crossover edge at the last stage survives 5 seeds ($+0.031 \pm 0.007$). A 5-seed verifier ablation shows the fixed mode's win is itself an artefact of the measured verifier and the 8-call cap: with an informative verifier and a cap the retries do not consume, verification helps ($+0.029 \pm 0.001$ quality for $+18\%$ cost) and `always_verify` becomes the right fixed mode. The gate reverts 32 of 40 candidates (23 false alarms on the quality criterion), and the AWR variant is promoted 14 times on a DR estimate biased by $+0.68$. The pattern across all six: the signal, not the learner, is the bottleneck.

1 Introduction

An agent harness is the scaffold around a model: the loop that decides which tool runs next, which model answers, whether a step is checked, when the episode ends. Inside that loop sit decisions that are made per request, in microseconds, by a rule somebody wrote: "always use the large model", "verify every sub-result", "refine if confidence is low". Each decision has a price in dollars and latency, and the rule does not see it.

Recent work treats the harness itself as the object of optimisation, at design time. The Darwin Gödel Machine evolves self-improving agents [43]; GEPA evolves prompts by reflection [2]; DarwinX [44], Meta-Harness [21], EnvHarness [12] and HarnessX [6] search over harness designs; EvoHarnessBench [16] grows a harness $H_t = \{\text{tools}, \text{skills}, \text{specialist agents}\}$ monotonically by stages and measures what a *fixed* agent policy forgets or gains per axis, reporting that agent-pool expansion alone gives -34.7% backward transfer. SkillOrchestra [40] routes each query to a typed agent in a pool by a learned competence profile with an explicit cost penalty, learned once from traces and then served statically.

This report studies the complementary problem. The harness is frozen; what moves is a small controller at its per-step decision points, trained from logged traffic, priced per request, and put into service only through a statistical gate. Three artefacts make up the method:

- **A priced controller.** A policy over (op, tier) pairs whose objective is quality minus a learned price of cost, trained offline by constrained group-relative policy optimisation on replay (or advantage-weighted regression with inverse-propensity weights), with the Lagrange multiplier found by damped dual ascent against a per-task budget and the served policy the iterate average.
- **An off-policy promotion gate.** Every served decision logs its propensity, exploration included, so a candidate policy can be scored on logged traffic by IPS, SNIPS and doubly-robust estimators with percentile-bootstrap confidence intervals, and is promoted only when the DR lower quality bound and the DR upper cost bound clear two bars.
- **Drift metrics.** Priced backward and forward transfer in the sense of Lopez-Paz and Ranzato [23], applied to the priced objective of the controller as the harness is repriced, re-versioned or grown.

Six experiments exercise these pieces, in increasing realism: a simulator pipeline (gate, drift, retrain); a counterfactual collection on CRMArena-Pro [13]; the gate's sample-size behaviour on that log; a 1.7B-parameter LLM as the controller; process-verified multi-agent orchestration; priced Socratic self-refinement; and the centrepiece, a continually trained orchestrator over a frozen multi-agent structure under scripted harness stages. Several results are negative, and they are reported as results: they say where the signal needed by a learned controller is and is not present, which is the question that decides whether a runtime controller earns its place at all. Every number in this report is taken from the committed result files of the public repository; nothing is re-estimated here.

2 Problem setting

The harness as a frozen decision process. A harness exposes, at each step t of an episode on task x, an observable state s_t and a legal set $\mathcal{A}(s_t)$ of actions $a = (\text{op}, \text{tier})$: an operation the harness can run (answer, retrieve, verify, escalate, delegate, re-plan, call a tool, engage a skill) on a model tier (cheap, mid, frontier). The harness's own transition $s_{t+1} \sim P_H(\cdot \mid s_t, a_t)$ is fixed; the controller $\pi(a \mid s)$ only chooses among what is legal. Each step has a realised cost c_t (tokens $\times$ list price) and latency ℓ_t; a terminal quality $q \in [0,1]$ is the harness's own score. We write $C = \sum_t c_t$ and $L = \sum_t \ell_t$. The harness is a frozen MDP over the controller's decision points, and the controller is the only thing trained.

Priced objective. The product specification is a budget, so the objective is

$$J(\pi) = \mathbb{E}_\pi[q] - \lambda\,\mathbb{E}_\pi[C] - \mu\,\mathbb{E}_\pi[L], \tag{1}$$

with $\mu = 0$ in every experiment here (latency is logged but not priced), and the constrained form

$$\max_\pi \ \mathbb{E}_\pi[q] \quad \text{s.t.} \quad \mathbb{E}_\pi[C] \le B, \qquad \mathcal{L}(\pi,\lambda) = \mathbb{E}_\pi[q] - \lambda\big(\mathbb{E}_\pi[C] - B\big),\ \lambda \ge 0. \tag{2}$$

The price λ is learned by projected dual ascent on an exponential moving average $\hat{c}$ of realised episode cost, with a proportional damping term (the PID-Lagrangian of Stooke et al. [33] without the derivative term; cf. RCPO [36]):

$$\hat{c}_k = \rho\,\hat{c}_{k-1} + (1-\rho)\,\bar{C}_k, \qquad \lambda_k^{\text{int}} = \Big[\lambda_{k-1}^{\text{int}} + \eta\,(\hat{c}_k - B)\Big]_+, \qquad \lambda_k = \Big[\lambda_k^{\text{int}} + k_p\,(\hat{c}_k - B)\Big]_+ . \tag{3}$$

Primal–dual iterates orbit the budget rather than converging to it, so the served policy is the iterate average $\bar{\pi}(a \mid s) = \frac{1}{|\mathcal{S}|}\sum_{W\in\mathcal{S}} \pi_W(a \mid s)$ over weight snapshots $\mathcal{S}$ from the second half of training [25], and $\bar{\lambda}$ is the mean price over the same half.

Policy class and logged propensities. The linear controller is a softmax over soft-clipped logits, mixed with ϵ-exploration:

$$z = L\tanh\big(Wx/L\big), \qquad \pi_\epsilon(a \mid x) = (1-\epsilon)\,\text{softmax}_{\mathcal{A}}(z)_a + \frac{\epsilon}{|\mathcal{A}|}, \tag{4}$$

where $x = \phi(s)$ is a feature vector and the softmax is over the legal set. The logit clip keeps the softmax steerable by λ and keeps importance ratios bounded. The propensity written to the log is $\mu_t = \pi_\epsilon(a_t \mid x_t)$, exploration included, so every served log is usable for off-policy evaluation (the logging discipline of Bottou et al. [4], Li et al. [22]).

Constrained GRPO on replay. When the harness can be re-rolled from a logged state (a simulator or a counterfactual lookup table), training samples a logged state s, draws a group of G continuations from the policy *as served* (π_ϵ, so the cost the dual variable tracks is the served cost), scores each by $R_i = q_i - \lambda C_i$ and uses the group-relative advantage with the group mean as baseline [31]:

$$A_i = R_i - \frac{1}{G}\sum_{j=1}^{G} R_j, \qquad \nabla_W \mathcal{L} = \sum_i A_i \sum_t \nabla_W \log \pi_\epsilon(a_t^i \mid x_t^i), \tag{5}$$

$$\nabla \log \pi_\epsilon(a \mid x) = \frac{(1-\epsilon)\,\pi(a \mid x)}{\pi_\epsilon(a \mid x)}\,\nabla \log \pi(a \mid x). \tag{6}$$

The last factor is the score correction for the ϵ-mixture: the gradient of the mixed policy is the bare softmax's score scaled by the share of the served probability that the softmax contributes.

Advantage-weighted regression with inverse propensities. When no replay exists, the trainer is AWR [27] on logged steps with return $R_i = q_i - \lambda\, C_i^{\rightarrow}$ (cost-to-go from the step), a per-bucket value baseline $\hat{V}$, and weights corrected by the logged propensity and self-normalised over the batch:

$$A_i = R_i - \hat{V}(b(x_i)), \qquad w_i = \frac{\exp(A_i/\beta)/\mu_i}{\sum_j \exp(A_j/\beta)/\mu_j}, \qquad \max_W \sum_i w_i \log \pi(a_i \mid x_i). \tag{7}$$

Here λ is a fixed price. Logged counterfactual arms, when present, are added as extra records (offline RL can only reweight what the log contains).

Off-policy estimators. For an episode i with per-step rewards r_t^i (the terminal quality, or the per-step cost) and cumulative ratio $\rho_{0:t}^i = \prod_{k\le t} \pi(a_k^i \mid s_k^i)/\mu_k^i$ clipped at $\rho_{\max} = 50$,

$$\hat{V}_{\mathrm{IPS}} = \frac{1}{n}\sum_i \sum_t \rho_{0:t}^i\, r_t^i, \qquad \hat{V}_{\mathrm{SNIPS}} = \sum_t \frac{\sum_i \rho_{0:t}^i\, r_t^i}{\sum_i \rho_{0:t}^i}, \tag{8}$$

$$\hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \Big[\hat{V}(s_0^i) + \sum_t \rho_{0:t}^i \big(r_t^i + \hat{V}(s_{t+1}^i) - \hat{Q}(s_t^i, a_t^i)\big)\Big], \qquad \hat{V}(s) = \sum_a \pi(a \mid s)\, \hat{Q}(s,a), \tag{9}$$

the per-decision importance-sampling estimator of Precup et al. [28], the self-normalised estimator of Swaminathan and Joachims [34], and the sequential doubly-robust estimator of Jiang and Li [14] (see also 37). The direct model $\hat{Q}$ is a per-feature-bucket mean fitted from the log and iterated under the candidate, i.e. bucketed fitted-Q evaluation [20]; the clip biases DR only through the direct-model residual. On a first-step contextual bandit (one decision per task, fixed continuation) the bandit form of Dudík et al. [8] applies with a *cross-fitted* direct model:

$$\hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \Big[\sum_a \pi(a \mid x_i)\, \hat{Q}^{(-i)}(x_i, a) + w_i\big(r_i - \hat{Q}^{(-i)}(x_i, a_i)\big)\Big], \qquad w_i = \frac{\pi(a_i \mid x_i)}{\mu_i}, \tag{10}$$

where $\hat{Q}^{(-i)}$ is fitted leaving out task i (the cross-fitting argument of Chernozhukov et al. [7]: an in-sample direct model shares the logged reward it corrects and shrinks the correction toward zero at small n). For multi-step orchestration logs with 2–4 decisions over 12 actions, a per-decision variant treats each logged decision as a bandit sample with single-step ratio $w_t = \pi(a_t \mid x_t)/\mu_t$ and reward the episode's quality or cost-to-go; it values changing one decision while the rest of the episode stays as logged, trading bias for ratios bounded by $|\mathcal{A}|/\epsilon$. Every estimate is reported with its effective sample size $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ [17].

The gate. Confidence intervals are percentile bootstraps over episodes (over tasks, with all logged arms of a task together, in the bandit case), 200 resamples, refitting the direct model on every resample so the DR interval includes direct-model variance. The decision is

$$\textsc{promote} \iff \hat{q}_{\mathrm{DR}}^{\mathrm{lo}} \;\ge\; q_{\mathrm{prev}} - \epsilon_q \quad \text{and} \quad \hat{c}_{\mathrm{DR}}^{\mathrm{hi}} \;\le\; B + \mathrm{tol}, \tag{11}$$

with q_{prev} the incumbent's quality: its on-policy log mean if it is the behaviour policy, its own DR estimate on the log otherwise, or the DR quality recorded at the incumbent's own promotion after a repricing (an over-budget incumbent would otherwise set a bar no budget-respecting candidate can pass). Defaults are $\epsilon_q = 0.03$ and $\mathrm{tol} = \$0.005$ (sim units 0.02).

Priced transfer. Let $R_{i,j}$ be the held-out priced value J_j of the policy in service at the end of stage i, evaluated on stage j, with J_j priced at the stage's own $\bar{\lambda}_j$. Following Lopez-Paz and Ranzato [23],

$$\mathrm{BWT}_j = R_{T,j} - R_{j,j}\ (j < T), \qquad \mathrm{FWT}_j = R_{j-1,j} - b_j, \tag{12}$$

with b_j the value of a uniform policy on stage j. For the two-version case, priced-BWT $= J(\text{frozen on } v_2) - J(\text{frozen on } v_1)$ and recovery $= J(\text{served on } v_2 \text{ after retrain and gate}) - J(\text{frozen on } v_2)$.

3 Platform

Harness interface. A host harness implements `reset`, `legal_actions`, `step` and `quality`; a simulable one also implements `snapshot` and `restore`, which is what constrained GRPO on replay needs. Its operations are the ops, its model pool the tiers, its version string a feature.

Trace schema. One record per controller decision: features, action, propensity (exploration included), the legal set the behaviour chose from, realised cost and latency, and, when the harness is simulable, a state snapshot for re-rolls. One episode per task carries the terminal quality and metadata (the agent's answer, the queries, the evaluator's calls).

Fail-closed spend ledger. Every priced call is appended to a ledger with its model, tokens, dollars, task and purpose. Before each call the ledger prices that call's worst case (input at 2 characters per token plus the full output cap) and raises *before sending* if spent plus worst case would exceed the cap; the collector flushes and exits with a distinct code. Realised spend therefore never exceeds the cap; a call that errors after being sent is charged at its worst case. The ledger is cumulative across runs and collection is resumable per (task, arm).

Simulator. Tasks have a latent difficulty (easy with probability 0.65, hard 0.35) that the controller never sees; it sees a proxy that is correct 85% of the time. Twelve actions, $\{\text{ANSWER}, \text{RETRIEVE}, \text{VERIFY}, \text{ESCALATE}\} \times \{\text{cheap}, \text{mid}, \text{frontier}\}$, each with its own cost, latency and quality tables; at most four steps, terminal ops only on the last. Version v_2 (the drift event) adds CACHED_RETRIEVE, near-free evidence that only works on easy tasks; ages the verifier so that VERIFY adds nothing on hard tasks; and reprices the frontier tier $\times 1.5$. Costs are in simulator units.

CRMArena-Pro host. CRMArena-Pro [13] is the first real host harness. Its ReAct agent [42] runs one closed loop with one fixed model and no per-turn hook, so the harness re-expresses that loop as `step()` and reuses everything else unchanged: task records, prompts, schema rendering, the live CRM connector and the benchmark's own evaluator (exact match, F1 for fuzzy categories, a privacy judge). Actions are `CONTINUE@tier` (one ReAct turn on that tier) and `ANSWER@tier` (force the final answer now); only `ANSWER@*` is legal at the 12-turn cap. Tiers are `claude-haiku-4-5` (cheap), `claude-sonnet-5-5` (mid) and `claude-opus-5-5` (frontier). Features are 43-dimensional: a bias, a one-hot over the 22 task categories, org type, turn one-hot and fraction, tool calls so far, cost so far, whether the last query returned rows, came back empty or errored, and a five-bucket context size. The evaluator's extractor and judge run on the cheap tier through the same capped path, are billed to the ledger, and are not charged to the controller's episode cost. Deliberate deviations from the benchmark's run script: no sampling parameters are sent (the mid and frontier models rejected them), calls are capped in tokens and time, observations are truncated to 20,000 characters, and a missing import in the benchmark's extractor fallback is supplied.

4 Experiments

4.1 Simulator pipeline: gate, drift, gated retrain

Setup. 3,000 episodes on v_1 under a uniform behaviour policy; constrained GRPO on replay at budget $B = 0.30$ (learned $\bar{\lambda} = 0.343$); gate the candidate against the behaviour log. Then drift: v_2 plus a scripted reprice(mid, 1.2); 6,000 fresh v_2 episodes served by the frozen policy with $\epsilon = 0.3$; retrain warm-started from the frozen policy; gate the adapted policy against the frozen one with the quality bar taken from the frozen policy's v_1 promotion. Ground truth is Monte Carlo over 20,000 fresh tasks with expected success probability. One seed per stage.

Results. Figure 1 shows both gates. On v_1, DR puts the candidate at quality 0.851 $[0.815, 0.887]$ and cost 0.306 $[0.302, 0.309]$; the decision is PROMOTE and the ground truth (0.880 at 0.303) is inside the DR interval. IPS and SNIPS sit low (0.742 and 0.761): the uniform behaviour over 12 actions and up to 4 steps gives cumulative ratios up to 326 against a near-deterministic candidate, which the clip at 50 truncates. After the drift, the adapted policy is scored at 0.873 $[0.854, 0.889]$ and cost 0.310 $[0.309, 0.311]$ against the frozen incumbent's 0.860 at 0.408; PROMOTE (upper cost $0.311 \leq B + \text{tol} = 0.320$). Table 1 gives the drift table: priced-BWT is -0.045 and recovery $+0.033$. Drift hits cost, not quality: the frozen controller keeps paying for an aged verifier and a repriced frontier, and the gated update restores the budget at equal quality.

Table 1: Simulator drift table (ground truth, $\lambda = 0.343$). priced-BWT $= J(\text{ii}) - J(\text{i}) = -0.045$; recovery $= J(\text{iii}) - J(\text{ii}) = +0.033$.

condition	quality	cost	within budget	J
(i) frozen on v_1	0.880	0.303	yes	0.776
(ii) frozen on v_2	0.870	0.406	no	0.731
(iii) adapted on v_2 (after gate)	0.871	0.312	yes	0.764

Limitations. The simulator's difficulty structure, success tables and drift event are hand-written; the run demonstrates the mechanics, not effect sizes. Single seed. The sim gate fits its direct model in-sample (buckets hold hundreds of transitions).

4.2 CRMArena-Pro counterfactual collection

Setup. 50 b2b tasks (2–3 per category, all 22 categories), each run on all three tiers with a fixed continuation (`CONTINUE` on that tier until the agent answers, `ANSWER` forced at the cap), so each task's first-step tier is a contextual bandit with a fully observed reward table and the held-out value of any tier policy is an exact lookup. The log holds 150 episodes from 832 priced calls, $17.98 under a $21 cap; episodes average 5.0 turns and 16 of 150 hit the 12-turn cap. The behaviour is

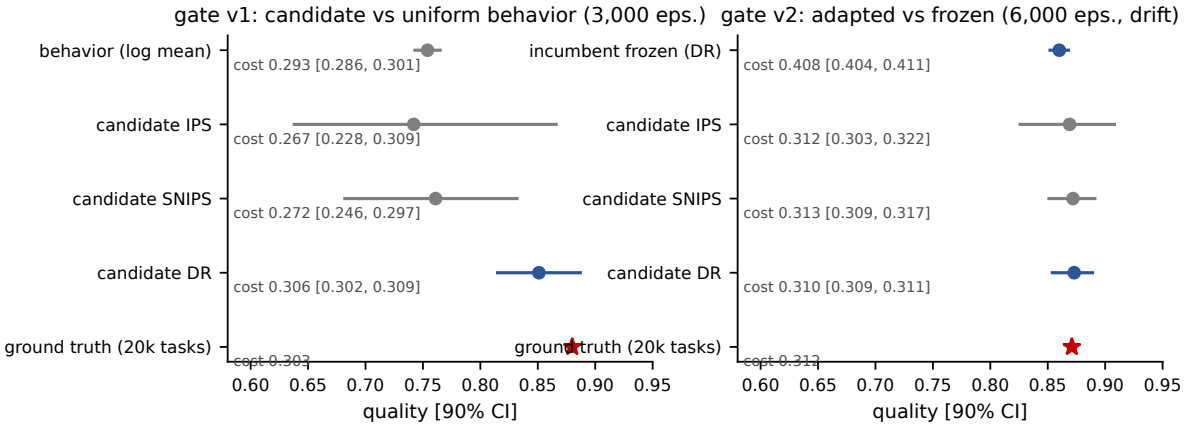


Figure 1: Simulator gates. Left: the constrained-GRPO candidate scored against 3,000 uniform-behaviour episodes on v_1; right: the adapted policy scored against 6,000 v_2 episodes served by the frozen policy. Points are estimates with 90% bootstrap CIs; the star is Monte Carlo ground truth. The gate decides on DR (blue). Values from the repository README (the run's JSON outputs are not committed).

uniform over tiers (propensity 1/3 at step 0, 1 after). The learned decision is the first-step tier from the 43-dimensional step-0 features. Policies are trained under 5-fold category-stratified CV over task ids (train on 40, evaluate on the held-out 10, seed 0), at fixed $\lambda \in \{2, 5, 10\}$ per dollar with AWR+IPS, and under a budget $B = \$0.10$ with constrained GRPO on the lookup table (learned λ).

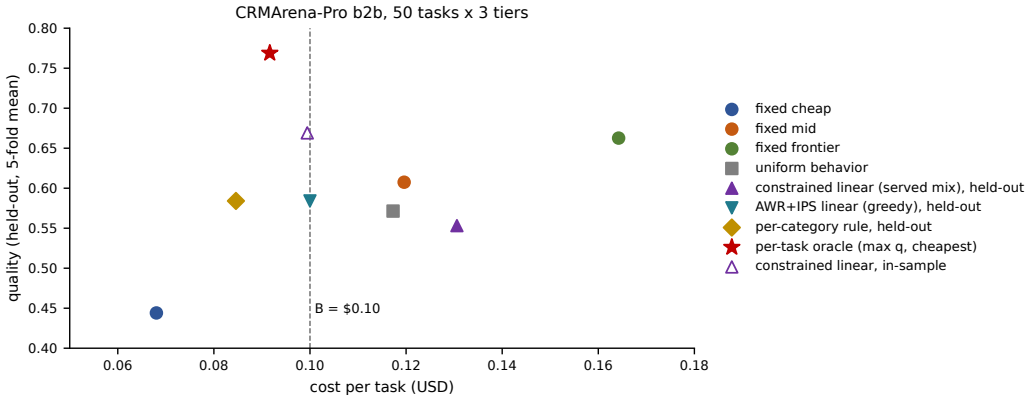


Figure 2: CRMArena-Pro b2b: held-out quality against cost per task (5-fold means, exact lookups). The per-task oracle beats every fixed arm on both axes; no learned policy does. The open triangle is the constrained policy in-sample (0.669 at \$0.099), the filled one held-out (0.553 at \$0.131).

Table 2: Held-out first-step tier policies on CRMArena-Pro b2b (5-fold means; cost in USD per task; across-fold SD of J in brackets, 0.05–0.3 throughout, as large as the gaps between rows).

policy (held-out)	$\lambda = 2$: J	$\lambda = 5$: J	$\lambda = 10$: J	$B = \$0.10$: quality / cost
AWR+IPS linear, served (stochastic)	0.350 [0.047]	0.045 [0.126]	−0.414 [0.246]	–
AWR+IPS linear, greedy	0.384 [0.076]	0.186 [0.126]	−0.273 [0.272]	–
per-category plug-in rule	0.415 [0.066]	0.161 [0.104]	−0.214 [0.154]	–
constrained cgrpo-lookup (iterate mix)	–	–	–	0.553 / 0.131 (over)
uniform behaviour	0.337 [0.066]	−0.015 [0.164]	−0.601 [0.340]	0.571 / 0.117
fixed cheap	0.308 [0.179]	0.104 [0.227]	−0.236 [0.308]	0.444 / 0.068
fixed mid	0.368 [0.092]	0.010 [0.124]	−0.588 [0.298]	0.608 / 0.120
fixed frontier	0.334 [0.133]	−0.159 [0.317]	−0.980 [0.633]	0.663 / 0.164
per-task oracle	0.639 [0.056]	0.452 [0.080]	0.148 [0.168]	0.769 / 0.092

Results. Table 2 and Figure 2. No learned linear rule clearly beats the best fixed arm held-out: the served AWR policy has lower J than the best fixed arm at every λ (0.350 vs mid 0.368; 0.045 vs cheap 0.104; −0.414 vs cheap −0.236), and its greedy version and the per-category rule beat the best arm by at most 0.08, inside the fold SDs. The features memorise: AWR's J at $\lambda = 2$ is 0.475 in-sample against 0.350 held-out; the constrained policy meets the budget in-sample (0.669 at

$0.099) and overspends held-out (0.553 at $0.131), where it is dominated by the uniform behaviour (0.571 at $0.117). Even the context-free arm mixture tuned to B on 40 tasks costs $0.115 on the held-out 10. The per-task oracle reaches 0.769 at $0.092, beating every arm on both quality and cost: the headroom is per task, not per category, and the missing piece is an observable that predicts which tier a task needs. Repricing events (frontier ×1.5, cheap ×3), with costs recomputed from logged tokens, push the frozen all-50 policy ($\bar{\lambda} = 1.36$/USD) over budget by +$0.029 and +$0.035: priced-BWT −0.040 and −0.048, in-sample recovery +0.069 and +0.091; but held-out the retrained policies still overspend (about $0.125) and the gate on the repriced realised log REVERTs all three retrains, which the held-out table says is the right call (recovery −0.012 for frontier ×1.5).

Limitations. Fifty tasks, one org, one seed; one task moves a fold's quality by 0.1. Only the first-step tier is learned, because the continuation was fixed at collection. The log holds one harness version, so version drift cannot be measured on it. Prompt caching was not used, so the roughly 5k-token system prompt is paid on every turn.

4.3 The gate's sample-size result

Setup. The candidate is the constrained policy trained on all 50 tasks (exact value: quality 0.665, cost $0.0987, within $B + \text{tol} = \$0.105$). The realised behaviour log is one uniformly drawn arm per task (50 episodes). The gate (cross-fitted DR, $\epsilon_q = 0.03$) is run once, then over 200 independent redraws of the realised log with a 200-resample bootstrap in each.

Results. On the single realised log DR estimates 0.740 [0.627, 0.855] and cost $0.0960 [0.0750, 0.1200]; the quality criterion passes and the cost criterion fails ($0.1200 > 0.105$): REVERT. On the full counterfactual log all three estimators reduce to the exact value (the weights over a task's three arms sum to 3) and the CI is pure task-sampling uncertainty from 50 tasks; the cost upper bound is still $0.1122 and the decision is still REVERT. Over the 200 redraws (Table 3, Figure 3) DR halves IPS's quality RMSE (0.048 vs 0.094) with 99% coverage of its 90% CI, but its cost CI is $0.046 wide on average, so the candidate, whose exact cost is $0.0013 under $B + \text{tol}$, is PROMOTEd in only 12/200 = 6% of redraws. The gate errs on the safe side; buying power means more tasks or a wider tolerance. As a back-of-envelope figure (not a measurement), with CI width scaling as $1/\sqrt{n}$, certifying a $0.005 margin needs a CI near $0.01, i.e. roughly 20× the tasks, about 1,000.

Table 3: Estimators over 200 redraws of the realised CRMArena-Pro log ($n = 50$, one arm per task). Cross-fitted direct model. PROMOTE in 12/200 redraws (6%).

metric	estimator	bias	RMSE	mean 90% CI width	CI covers exact
quality	IPS	+0.011	0.094	0.338	94%
quality	SNIPS	+0.001	0.048	0.238	100%
quality	DR	+0.004	0.048	0.256	99%
cost	IPS	+0.0001	0.0114	0.0414	92%
cost	SNIPS	−0.0009	0.0102	0.0381	92%
cost	DR	−0.0018	0.0114	0.0461	92%

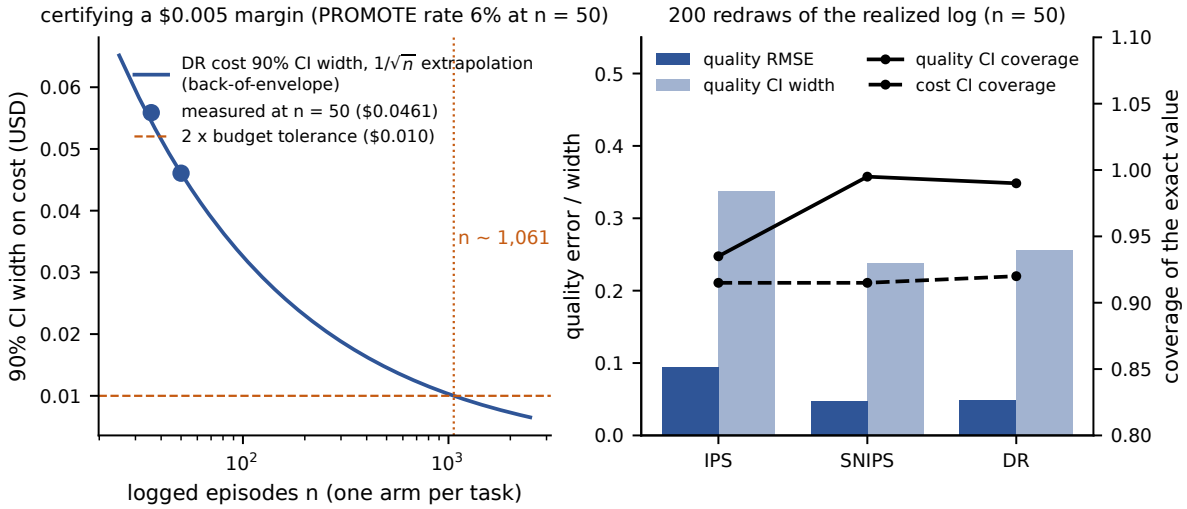


Figure 3: Gate power. Left: the measured DR cost CI width at $n = 50$ and its $1/\sqrt{n}$ extrapolation (a back-of-envelope, not a measurement) against twice the budget tolerance. Right: estimator RMSE, CI width and coverage over the 200 redraws.

Limitations. The in-sample direct model covers 96%/94% with a −0.006 quality bias, a difference inside the noise at $n = 50$; the redraw study reports bias and coverage empirically rather than assuming them.

4.4 LLM controller: GRPO with LoRA on a 1.7B model

Setup. The controller is Qwen3-1.7B [30] with a LoRA adapter [11] ($r = 16$ on the attention and MLP projections, bf16), trained with GRPO. The agent models stay frozen; the tiers are the arms. The prompt shows the difficulty proxy, the state, per-tier prices, the budget and the legal actions, and the completion must be exactly one tag `<action>TIER OP</action>`; anything else is invalid (reward −1, cost booked as the cheapest safe action). After the first action a fixed continuation answers on the same tier; the reward is $q - \lambda C$ with λ learned by the same damped dual ascent, once per step from the batch mean cost. Each step generates 64 completions (8 prompts × 8 generations in phases 1 and 1b; 4 × 16 in 1c), length 16, lr 10^{-5}, over 2,000 seeded sim tasks with the legal-action order shuffled per task (with a fixed order the base model only ever sampled one or two arms). Three runs on the simulator, each run once on one A100-40GB: `phase1` (λ lr 0.1, k_p 2.0, β_{KL} 0.02, 300 steps, 1,422 s, about $0.83); `phase1b` (slower dual step: lr 0.02, k_p 0.5, EMA 0.95; β_{KL} 0.05; entropy 0.01); `phase1c` (150 SFT steps on per-difficulty oracle labels at a price shown in the prompt and drawn from $\{0.0, 0.1, \ldots, 2.0\}$, adapter merged so the KL reference is the SFT policy, then 200 GRPO steps with 16 generations, temperature 1.2, $\lambda_0 = 0.5$; 1,134 s of training plus 278 s of eval, about $0.82). A fourth run (`crm_phase2`) applies the same recipe to CRMArena-Pro fold 0 in lookup mode (574 s, about $0.35).

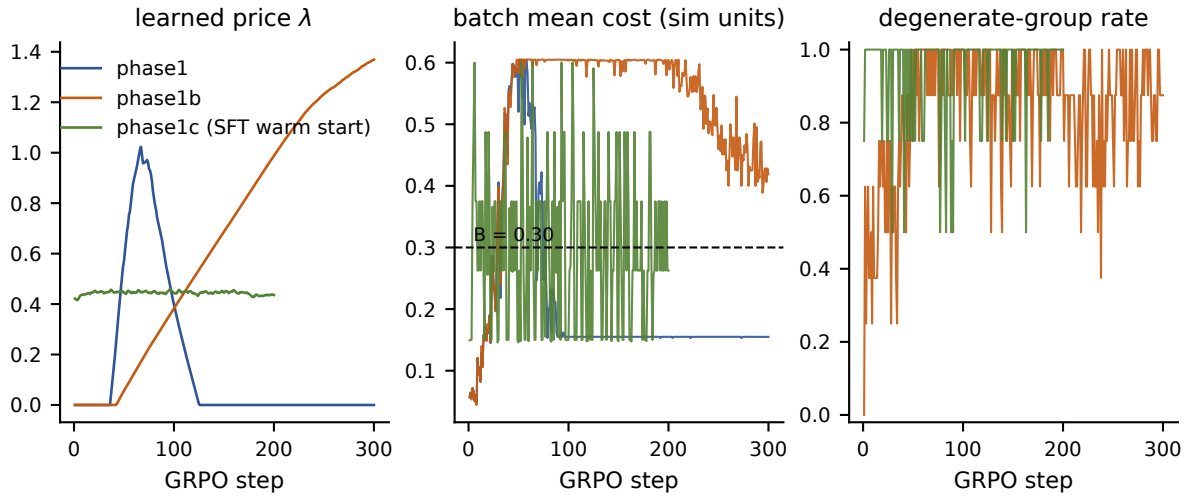


Figure 4: GRPO training logs. `phase1`'s price orbits once and decays to 0 as the policy collapses onto `mid RETRIEVE`; `phase1b`'s price is still rising at step 300 while cost stays above budget; `phase1c`'s price settles at $\lambda \approx 0.44$ with the batch cost alternating between two deterministic rules. Degenerate groups (all completions identical, hence zero advantage) dominate every run.

Table 4: LLM controller on sim v_1 (2,000 fresh tasks, seed 777; budget $B = 0.30$; the policy's first action then `ANSWER` on the same tier). "Within" means cost $\leq B + 0.02$.

policy	quality	cost	within	invalid
Qwen3-1.7B zero-shot (greedy, no adapter)	0.586	0.048	yes	18.9%
`phase1`, steps 100 / 200 / 300	0.748–0.749	0.155	yes	0%
`phase1b`, step 100	0.885	0.603	no	0%
`phase1b`, step 300	0.832	0.434	no	0.1%
`phase1c`: SFT only, and GRPO steps 100 and 200 (all identical)	0.848	0.330	no	0%
linear controller (`policy_v1_sim.json`), first action only	0.573	0.035	yes	0%
fixed `ANSWER` on cheap / mid / frontier	0.522 / 0.688 / 0.820	0.020 / 0.100 / 0.400	yes / yes / no	0%
best deterministic per-proxy rule (easy: mid ESCALATE; hard: frontier ANSWER)	0.814	0.250	yes	–
best randomised policy at cost exactly B (LP)	0.835	0.300	yes	–
derived: `phase1c` served with the price drawn as in training's last 100 steps	0.837	0.305	yes	–

Results. Table 4 and Figure 4. GRPO alone never found the budgeted optimum. *Group degeneracy*: within a prompt group the completions are almost always identical, 83.5% of groups on average in `phase1b`, which gives zero GRPO advantage, so the only learning signal is λ moving the mean reward between steps and the policy jumps between near-deterministic arms instead of learning the per-proxy split. *Primal–dual orbit*: in `phase1` the policy swings to `frontier RETRIEVE` (cost 0.60), λ climbs to 1.02 by step 67, the policy collapses onto `mid RETRIEVE` (0.155) and λ decays to 0 with nothing to pull it back; in `phase1b` λ is still rising at step 300 (1.37) and the policy has not reached the budget. Mixing is not the bottleneck: a deterministic per-proxy rule reaches 0.814 at 0.250 and randomisation adds only 0.021; what the runs failed to do is condition on the proxy. In `phase1c` the SFT stage reached 100% held-out agreement with the oracle

labels (1,000 prompts, 0% invalid) and splits by difficulty where the oracle does; GRPO then left the policy unchanged on all 2,000 eval tasks (the degenerate-group rate averaged 93% over steps 1–100 and 97% over 101–200 even after the rollout temperature was raised from 1.2 to 1.5 at step 20). GRPO's contribution is the price: λ settles at 0.44 (mean 0.443 over the last 100 steps), which is the slope at which the hard-proxy action flips. At that price the SFT policy ($J = 0.702$) ties the oracle rule (0.703) and the LP (0.702). No single greedy policy meets $B = 0.30$: at any one shown price the policy is a deterministic rule costing 0.33 or 0.15; training met the budget by alternating the shown price over time (batch cost 0.301 and 0.296 over the two halves), and serving can reach it by drawing the shown price (the price-conditioned analogue of iterate averaging): 0.837 at 0.305, derived from the two greedy rules rather than decoded end to end. On CRMArena-Pro fold 0, the SFT policy ignores the shown price (the same per-category rule at every λ, about \$0.075, under budget), so λ fell to 0 by step 17; GRPO moved training-set quality from 0.705 to 0.730 and left the 10 held-out tasks unchanged at 0.700 / \$0.073, against fixed frontier 0.700 / \$0.142 and the fold-0 linear policy 0.590 / \$0.143. That is one 10-task fold where one task is 0.1 of quality and the 5-fold mean of the same category rule is 0.584 / \$0.085 (Table 2): a single draw, not evidence of generalisation or of the LLM beating the linear controller.

Limitations. Each GPU run ran once; no seed variance. The phase-1 table truncates the linear policy's continuation (its first action is `cheap RETRIEVE` because it was trained to keep going; with its own continuation it scored 0.876 at 0.301 in the run that produced that file), so its row is not its real performance. Showing λ rounded to 0.1 puts $\bar{\lambda} = 0.443$ on the over-budget side of the flip point.

4.5 Process-verified multi-agent orchestration

Setup. A planner (cheap tier) decomposes a CRM request into 1–3 sub-queries; a data worker (mid) runs one sub-query at a time through the harness's own step path; a verifier (cheap) checks a sub-result against its goal (a structural pre-check fails at \$0 on query errors); the answer is the harness's forced `ANSWER@mid`. After every worker step an orchestration policy chooses $\{\text{VERIFY}, \text{SKIP_VERIFY}, \text{REPLAN}, \text{ANSWER}\} \times$ tier; a fail retries the sub-query once; at most 8 role calls per episode. Fixed modes `always_verify` and `never_verify`, plus a uniform exploration mode with logged propensities, were run on the same 50 b2b tasks as the counterfactual log: 130 episodes (50 + 50 + 30, stopped by the ledger cap), \$10.97, \$0.084 per episode. An orchestration reward model (ORM: logistic regression and a 16-unit MLP over process features) is evaluated out of fold over tasks.

Table 5: Decomposition against the single ReAct agent on the same 50 CRMArena-Pro b2b tasks (single-agent rows are exact lookups of the counterfactual log). The six confidentiality tasks score 0 for every system, so the ceiling is 0.88.

system	accuracy / quality	cost / task	$J(\lambda = 2)$	$J(\lambda = 5)$
single agent, fixed cheap	0.444	\$0.0680	0.308	0.104
single agent, fixed mid	0.608	\$0.1196	0.368	0.010
single agent, fixed frontier	0.663	\$0.1642	0.334	−0.159
single agent, per-task oracle (max quality, cheapest)	0.769	\$0.0916	0.586	0.311
MAS `always_verify`	0.480	\$0.0906	0.299	0.027
MAS `never_verify`	0.461	\$0.0821	0.297	0.051
MAS `controller:uniform` (30 tasks, exploration)	0.535	\$0.0758	0.384	0.156

Paired, same 50 tasks: `always_verify` vs `never_verify`: 6 wrong→right, 5 right→wrong, net +1 ($\Delta q = +0.019 \pm 0.067$ SE, +\$0.0086). `always_verify` vs single mid: both right 19, MAS right / mid wrong 5, MAS wrong / mid right 11, both wrong 15 (net −6).

Table 6: Orchestration reward model, out of fold over tasks (130 episodes, base rate $P(\text{correct}) = 0.485$, base-rate Brier 0.250).

model	features	AUROC	Brier	ECE (5 bins)
logreg	process features (default)	0.469	0.281	0.114
logreg	process − verifier features	0.451	0.280	0.140
logreg	process + category one-hot	0.831	0.165	0.040
mlp	process features (default)	0.456	0.263	0.053
mlp	process + category one-hot	0.831	0.164	0.035
–	category alone, no process feature	0.821	0.166	–

Results. The headline is negative (Table 5): the decomposition scores below the single mid-tier agent (0.480 vs 0.608) at lower cost and is dominated on the priced frontier at both prices (by mid at $\lambda = 2$, by cheap at $\lambda = 5$); paired on the 50 tasks the single mid agent wins 11 and loses 5. This is the compute-matched single-agent comparison MAS-Orchestra [15] reports as the bar, and it is not cleared. Three mechanisms are visible in the log: the plan is fixed up front by a cheap model with no data in hand while the single agent picks each query after seeing the last result; `never_verify` gives the strong model about 2.7 worker turns and a forced answer against the single agent's about 5; and under `always_verify` 47 of 50 episodes hit the 8-call cap because 65% of verifier calls fail and each fail spends a retry. Verification bought +1 task for +10% cost, and the paired table says that is noise (6 flips each way, all through a fail and a retry). The verifier is an

alarm, not a judge: it flags 24 of 26 wrong episodes ($P(\text{flag} \mid \text{wrong}) = 0.92$, silent failures 2/50) and 23 of 24 correct ones ($P(\text{flag} \mid \text{correct}) = 0.96$), so $P(\text{wrong} \mid \text{flagged}) = 0.51$ against a base rate of 0.52. The ORM (Table 6) finds that process features carry no out-of-fold signal (AUROC 0.469, Brier above the base rate) and the task category carries most of it (0.821 alone; 0.831 with process features), with CV over tasks so that this is category-level difficulty transferring to unseen tasks, resting on 2–3 training tasks per category. The priced-controller hook was exercised, not evaluated: AWR on the 30-episode ORM-relabelled exploration log changes nothing detectable per decision (per-decision SNIPS 0.563 $[0.373, 0.701]$ against the behaviour's 0.567, ESS 55 of 64), and the episode-level estimate has ESS 7 of 30.

Limitations. The verifier is a cheap model of the same family checking a mid model; no human process label. The 8-call cap interacts with a high-fail verifier, so a higher cap or a better verifier is a different system. Sequential sub-queries in one growing context.

4.6 Priced Socratic self-refinement

Setup. One Socratic self-refinement round [32] on a CRMArena-Pro answer: a cheap probe decomposes the answer into 1–3 claims with self-reported confidences (about \$0.003, so a gate can look before it pays); each claim is re-solved by $k = 2$ worker mini-episodes on the chosen tier (≤ 3 turns, the SOQL runs on the real org); a rewrite on that tier produces the refined answer; the benchmark's evaluator scores it. The initial answers are the cheap-tier episodes of the counterfactual log (already scored and paid). The run logged a tree, a refinement on every tier for every task, so every policy over {STOP, REFINE@cheap, REFINE@mid} is an exact lookup: `never`, `always_1:t`, `uncertainty_gated:`τ`:t` (τ swept 0.3–0.9 on the minimum claim confidence), and a softmax-linear priced controller over 36 features trained by exact policy gradient on the tree (budget $B = \$0.10$ with learned λ, or fixed $\lambda \in \{2, 5\}$; 5-fold CV over tasks). The run stopped at a \$14 cap: 42 tasks probed, 41 complete on both tiers, \$13.97, \$0.333 per tree against a \$0.36 estimate. One task is 0.024 of quality.

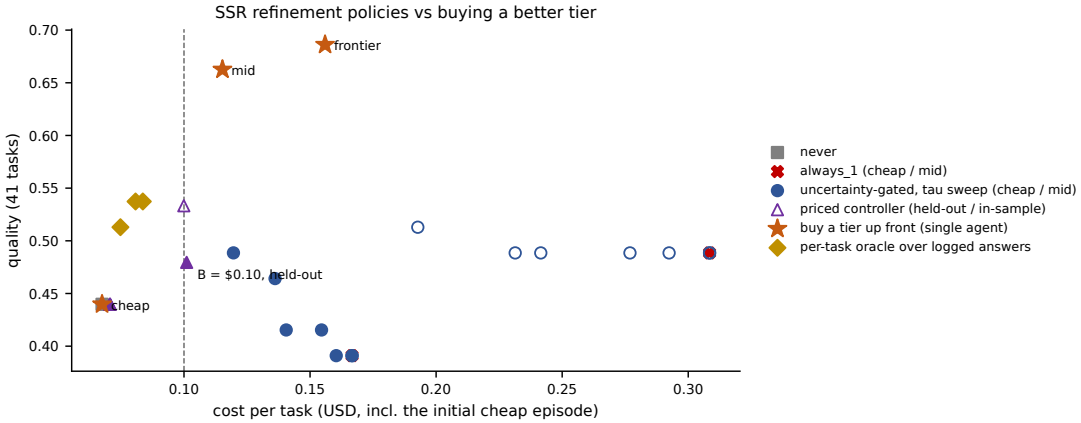


Figure 5: SSR policies on the logged tree (41 tasks, cost includes the initial cheap episode), from `frontier.csv`. Open circles are mid-tier gates, filled are cheap-tier. The single-agent mid arm at \$0.115 dominates every refinement policy.

Table 7: Refinement harm after one round (41 tasks). "fix" is wrong→right out of initially wrong, "harm" right→wrong out of initially correct.

tier	changed	fix	harm	mean Δq	round cost	worker turns
cheap	85% (35/41)	13% (3/23)	28% (5/18)	−0.049	\$0.0962	14.7
mid	44% (18/41)	13% (3/23)	6% (1/18)	+0.049	\$0.2380	13.9

Results. The probe's confidence is weakly informative: AUROC 0.65 (90% CI 0.51–0.79, $n = 41$) of $1 -$ min claim confidence for predicting a wrong initial answer; the decomposer is pessimistic about everything (minimum claim confidence never above 0.75, below 0.4 on 26 of 41 tasks whether right or wrong), and the re-solve signal (AUROC 0.67 cheap, 0.55 mid) is no better than the free one; 22% of mid worker samples are "None". Refinement harms as often as it helps (Table 7): a cheap round fixes 3 of 23 and breaks 5 of 18 ($\Delta q = -0.049 \pm 0.069$), a mid round fixes 3 and breaks 1 ($+0.049 \pm 0.049$), both within one SE of zero. Policies (Figure 5): `never` 0.440 ± 0.078 at \$0.068; `always_1:cheap` 0.391 at \$0.167; `always_1:mid` 0.489 at \$0.309; the best fixed cheap gate ($\tau = 0.3$) 0.489 at \$0.120; the budgeted controller 0.480 ± 0.077 at \$0.101 held-out (paired $+0.040 \pm 0.033$ over `never`, refining 20% of tasks, $\bar{\lambda} = 0.09$/USD because it sits under budget; 0.533 at \$0.100 in-sample), and the fixed-λ controllers learn `never`. Even the clairvoyant choice among the three logged answers per task reaches only 0.537 at \$0.084. H1 (refine where confidence is low) is not established; H2 (always-refine is dominated under a budget) is supported, on the cheap tier by `never` itself; H3 (a priced gate reaches the frontier) only in the weak sense that

the held-out controller sits on the frontier by learning to refine rarely. The comparison that matters is refine-the-cheap-answer against buy-mid-up-front, and buying the tier wins by a wide margin: the single-agent mid arm on the same 41 tasks scores 0.663 ± 0.074 at \$0.115; the best cheap-refinement policy at that price is -0.174 ± 0.085 paired for +\$0.004; `always_1:cheap` is -0.272 ± 0.099; even refining on mid at 2.7× the cost is -0.174 ± 0.092; frontier up front (0.686 at \$0.156) beats every refinement policy by 0.17–0.30. The sub-questions need the same schema reasoning that made the cheap agent fail, and the workers inherit it.

Limitations. One round, 41 tasks, one initial arm; the τ sweep is tuned on the tasks it is evaluated on. The in-sample controller row shows what 22 category one-hots memorise on 41 tasks. Agreement with $k = 2$ is weak, and no sampling parameters are sent.

4.7 The continual orchestrator

Harness definition. The multi-agent structure of Section 4.5 is expressed as a typed harness in the terms of SkillOrchestra and EvoHarnessBench: an agent pool of typed specialists (agent id, tier, mode, stage) with modes planner, data-worker, verifier and answer; tools the orchestrator can `call@`; skills it can `engage@`; and a competence handbook over the 22 task categories (the axis the real run found carries the signal). The harness $H_t = \{\text{tools}, \text{skills}, \text{agents}\}$ grows monotonically by stages, $H_1 \subseteq H_2 \subseteq \cdots$, plus a price axis; the structure at each stage is frozen and the *policy* at the decision point is what adapts, so the question is EvoHarnessBench's turned around: when H expands under an orchestrator that keeps learning, what happens to the policy's backward and forward transfer, and does the gate keep it safe without making it too slow to matter? Eighteen actions: `delegate@data-1..4`, `verify@verify-1,2`, `replan@plan-1..3`, `answer@answer-1..4`, `call@schema_lookup`, `call@aggregate_calc`, `engage@soql_aggregate`, `engage@date_window`, `engage@decompose_fine`; and 88 features (the real orchestrator's 66, computed by its own feature function on a view of the simulated state, plus availability flags, tools called, skills engaged, stage and a last-fail flag). Unavailable capabilities have fixed zero feature slots and zero logits, so a policy trained at stage g keeps its weights when stage $g + 1$ arrives.

Simulator and calibration. Days of traffic under several harness changes are beyond the real-run budget, so the logged process is re-expressed as a seeded simulator whose decision points, legal-set rules and feature layout are the real orchestrator's. Every parameter is either measured from the two committed logs (130 MAS episodes, 150 single-agent episodes), with its count, or marked assumed (Appendix B). Measured: plan-length distribution (4/22/104 of 130), worker step kinds per tier, the retry re-draw, the verifier's per-call fail rates by step kind (72 of 124 on rows, 23 of 24 on empty, 12 of 12 structural at \$0) and its episode-level $P(\text{flag} \mid \text{wrong}) = 0.92$ and $P(\text{flag} \mid \text{correct}) = 0.96$, token means per role and tier, per-category accuracy, and the configuration. Assumed: whether a rows step is the *right* rows ($P(\text{good} \mid \text{rows}) = 0.60/0.75/0.85$ by tier), the logit slope from plan coverage to the answer (1.5, centred so `never_verify` reproduces the category base rates), a task-level logit noise (sd 0.4), and everything that arrives after stage 0. On the stage-0 harness the two fixed modes reproduce the real run's process numbers (Table 8).

Table 8: Simulator calibration check: stage-0 harness, 1,000 simulated tasks per mode against the real run's 50 tasks per mode (sim / measured).

mode	accuracy	cost/task	role calls	verifier calls	cap hits	$P(\text{flag} \mid \text{wrong})$	$P(\text{flag} \mid \text{correct})$
`always_verify`	0.446 / 0.480	\$0.0894 / \$0.0906	7.87 / 7.88	2.94 / 2.94	95% / 94%	0.95 / 0.92	0.93 / 0.96
`never_verify`	0.459 / 0.460	\$0.0827 / \$0.0821	4.75 / 4.74	0 / 0	0% / 0%	–	–

Pre-registered stages. Both schedules were written down before the first full run and not changed afterwards (Table 9). The `mild` schedule keeps the days and axes with smaller effects: reprice mid ×0.8 / frontier ×1.2, `verify-2` at $P(\text{flag} \mid \text{correct}) = 0.60$, `data-4` at rows 0.78 / good 0.75, `answer-4` mid-grade on 3 categories, half the tool and skill bonuses, planner v2 at 1–3 sub-queries.

Table 9: The pre-registered default schedule: 40 days, events on days 9 / 17 / 25 / 33.

stage	day	axis	what changed
g0 baseline	1	–	the real run's pool: plan-1, data-1..3, verify-1, answer-1..3; list prices
g1 reprice	9	price	mid ×0.5, frontier ×1.5
g2 agents	17	agents	+verify-2 (a better verifier: $P(\text{flag} \mid \text{correct})$ 0.96 → 0.30 at the same recall), +data-4 (database specialist, cheap), +answer-4 (reporting specialist, cheap)
g3 tools	25	tools	+schema_lookup (the next worker step cannot error), +aggregate_calc
g4 skills	33	skills	+soql_aggregate, date_window, decompose_fine; the planner now decomposes into 1–4 sub-queries

The day loop and comparators. Each day is 96 seeded tasks, the same for every arm. The continual arm serves its incumbent with $\epsilon = 0.1$ and logs propensities; a learner trains every day by constrained GRPO on replay from the last 5 days of logged states (60 iterations, batch 16, group 6; λ learned against $B = \$0.090$ per task; iterate averaging), warm-started from itself; the gate (sequential DR with 200 bootstrap resamples, $\epsilon_q = 0.05$, tol $= \$0.005$) scores the learner against the incumbent on the log served since the last harness event (up to 5 days), with the budget net of the policy-independent prefix (planner plus first worker call, \$0.022 at list prices). An AWR+IPS variant (fixed $\lambda = 2$, 200 epochs) runs the same loop. Baselines on the same traffic: `frozen` (the learner as of day 5, served without ϵ, restricted to stage-0 actions); `so_static`, a SkillOrchestra-style router whose handbook is the literal rule competence$[(a, \text{category})] - \lambda_c \cdot \overline{\text{cost}}[a]$ with cost tie-break, learned from the warm-up exploration traces (days 1–4) with λ_c chosen on day 5 by episode-level SN-IPS, never updated (new capabilities scored at the handbook's prior); `so_relearn`, the same router re-fitted after every stage (3 days served with $\epsilon = 0.3$, then a refit and re-tune, the lag counted); `always_verify`; `never_verify`; and a per-stage `oracle` (constrained GRPO on 200 fresh exploration episodes of that stage, 300 iterations, no gate: a reference, not a proven optimum). Held-out: 150 fixed tasks per stage, simulator expected quality, never seen by the gate; $J_g = \text{quality} - \bar{\lambda}_g \cdot \text{cost}$ with $\bar{\lambda}_g$ the stage oracle's learned price (1.25 / 0.85 / 0.81 / 0.66 / 0.73 per USD for g0–g4), at which the budgeted optimum maximises the Lagrangian so J_g has a ceiling and regret is well defined. Metrics follow Eq. (12); in addition, hit_j is the event-day policy's J on stage j minus on stage $j-1$, $\text{recovery}_j = R_{j,j} - R_{j-1,j}$, days-to-recover is the first day of the stage within 0.02 of the oracle, stationary J the mean over the stage's last four days, and cumulative regret the sum over days 6–40 of the oracle's J minus the serving policy's on the day's stage. One seed; 189 s on CPU.

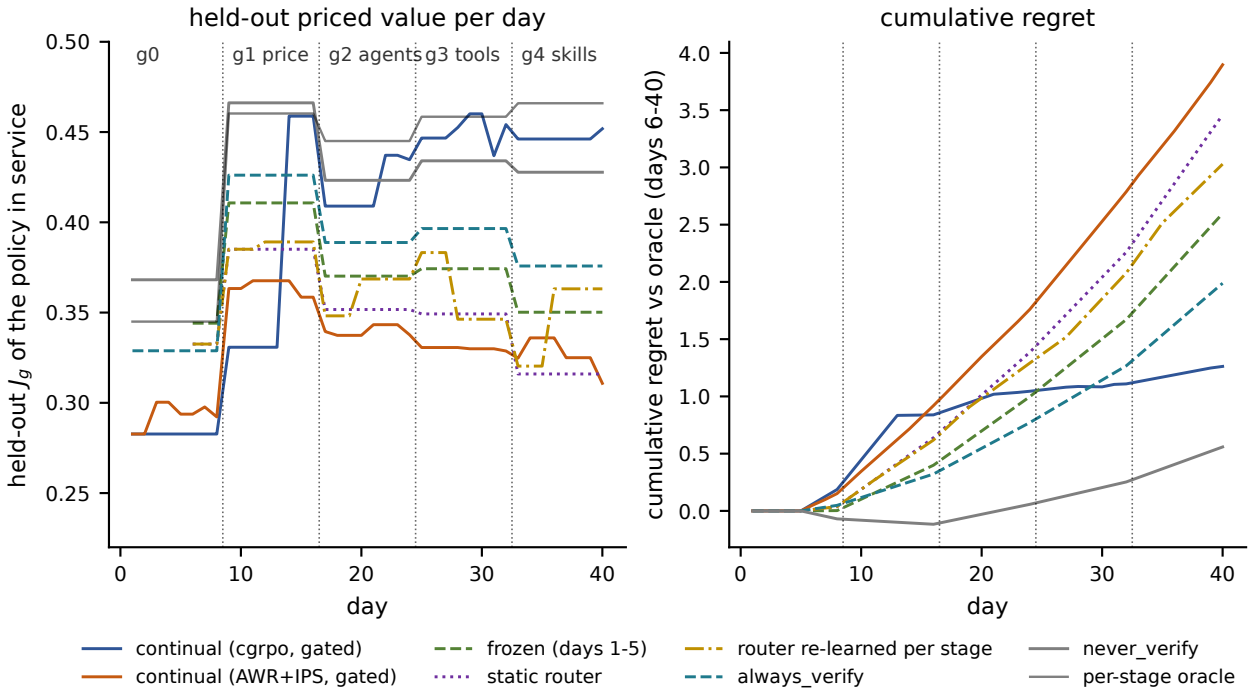


Figure 6: The continual orchestrator over 40 days (default schedule, from `days.jsonl`). Left: held-out J_g of the policy in service each day; the grey upper line is the per-stage oracle. Right: cumulative regret to the oracle from day 6. The continual (cgrpo) arm is the only learned arm that recovers after every stage; `never_verify` has the lowest regret of all.

Results, stated in order of importance. *(1) The fixed mode `never_verify` wins; against the learned alternatives the continual orchestrator wins on every metric.* Cumulative regret over days 6–40 (Table 11, Figure 6): `never_verify` 0.56, continual (cgrpo, gated) 1.26, `always_verify` 1.99, frozen 2.60, router re-learned per stage 3.03, static router 3.46, continual AWR 3.90. `never_verify`, the all-mid fixed mode, is at the oracle on g1 (0.466 vs 0.460) and within 0.02–0.04 on g2–g4 without learning anything: after the mid repricing, running every sub-query and the answer on the half-price mid tier is the budgeted optimum or near it in this simulator. The continual arm ends every stage within 0.001–0.014 of the oracle and is the only learned arm that recovers at all (days to recover 6 / 6 / 1 / 1); the frozen controller, the static router and the re-learned router never come within 0.02 of the oracle on any stage. The continual arm beats `never_verify` only in the stationary periods of the capability-adding stages (g2 0.429 vs 0.423, g3 0.453 vs 0.434, g4 0.448 vs 0.428) and loses that back during the recovery days. Over 5 seeds of the same configuration (Section 4.7, variant A; mean ± SE), the continual-minus-`never_verify` stationary J is -0.050 ± 0.011 (g0), -0.009 ± 0.002 (g1), -0.014 ± 0.007 (g2), $+0.011 \pm 0.010$ (g3) and $+0.031 \pm 0.007$ (g4), and the whole-stream regret is continual 1.06 ± 0.21, `never_verify` 0.74 ± 0.08, `always_verify` 1.97 ± 0.09: the fixed mode's whole-stream win and the learner's margin at the last stage are both outside seed noise; the g2–g3 margins are not. The honest answer to "does the learner beat the alternatives" is: it beats every design-time learner,

and the thing that beats it is knowing the right fixed policy in advance, which the growing pool makes less likely with every stage (the fixed mode's gap to the oracle grows from −0.006 at g1 to −0.038 at g4).

Table 10: Per-event transfer on the default schedule (held-out J of the policy in service; oracle J per stage g1–g4: 0.460 / 0.445 / 0.458 / 0.466; uniform 0.351 / 0.324 / 0.317 / 0.318). Stationary J per stage and BWT per stage are in Appendix C and Figure 9.

arm	hit at event g1/g2/g3/g4	FWT g1/g2/g3/g4	days to recover	gap to oracle at stage end g1/g2/g3/g4
continual (cgrpo, gated)	+.048/ − .050/ + .012/ − .008	−.020/ + .085/ + .130/ + .128	6/6/1/1	−.001/ − .010/ − .004/ − .014
continual (AWR+IPS, gated)	+.071/ − .019/ − .007/ − .004	+.012/ + .016/ + .014/ + .006	–/–/–/–	−.102/ − .107/ − .130/ − .155
frozen (days 1–5)	+.067/ − .041/ + .004/ − .024	+.060/ + .046/ + .057/ + .032	–/–/–/–	−.050/ − .075/ − .084/ − .116
static router	+.053/ − .033/ − .002/ − .033	+.034/ + .028/ + .032/ − .002	–/–/–/–	−.075/ − .093/ − .109/ − .150
router re-learned per stage	+.053/ − .041/ + .015/ − .026	+.034/ + .024/ + .066/ + .002	–/–/–/–	−.071/ − .076/ − .112/ − .103
`always_verify`	+.097/ − .037/ + .008/ − .021	+.075/ + .065/ + .080/ + .057	–/–/–/–	−.034/ − .056/ − .062/ − .090
`never_verify`	+.098/ − .043/ + .011/ − .006	+.115/ + .099/ + .117/ + .109	1/–/–/–	+.006/ − .022/ − .024/ − .038

Table 11: Whole-stream summary, default schedule (mild schedule in parentheses).

arm	mean BWT	mean FWT	regret, days 6–40
continual (cgrpo on replay, gated, learned λ)	+0.031 (+0.035)	+0.081 (+0.075)	1.262 (1.100)
continual (AWR+IPS, gated, $\lambda = 2$)	−0.006 (−0.005)	+0.012 (+0.002)	3.896 (3.668)
frozen (trained days 1–5)	0	+0.049 (+0.053)	2.597 (1.970)
SkillOrchestra-style static router	0	+0.023 (+0.025)	3.459 (2.902)
SkillOrchestra-style router, re-learned per stage	+0.019 (+0.017)	+0.032 (+0.014)	3.029 (3.057)
`always_verify`	0	+0.069 (+0.062)	1.988 (1.731)
`never_verify`	0	+0.110 (+0.100)	0.559 (0.394)

(2) The whole-stream headline is a cold-start artefact: the learner crosses ***`never_verify`*** *around day 22–25 and wins every stage window after that, while* ***`never_verify`*** *falls further below the oracle at each capability stage.* Figure 7. Served J, continual minus `never_verify`, by 8-day stage window: g0 −0.073, g1 −0.070, g2 −0.021, g3 +0.024, g4 +0.035; on the held-out sets −0.085, −0.087, −0.004, +0.016, +0.019. The served gap is positive from day 24 on and stays positive. Cumulative held-out regret to the oracle restricted to days 17–40 is 0.42 (continual) against 0.68 (`never_verify`); restricted to days 25–40 it is 0.22 against 0.50, and every other arm is further behind (frozen 1.60, re-learned router 1.74, static 2.07, AWR 2.14 from day 25). The learner's served J includes its $\epsilon = 0.1$ exploration mass, a tax the fixed policy does not pay, and it still wins after the crossover. The whole deficit in the whole-stream number is the 12 days on the uniform policy before the first promotion (finding 3) plus the repricing recovery (days 9–13, where the half-price mid tier makes `never_verify` the optimum before the learner has moved): `never_verify` is above the reference oracle at g0 and g1 (+0.023, +0.006 on the held-out sets at stage end) and falls to −0.022, −0.024, −0.038 at g2–g4 as capabilities it cannot use arrive. The post-crossover margin at g4 holds over 5 seeds (+0.031 ± 0.007, above); the stage-2+ capabilities are assumed parameters (Appendix B), so its size is the simulator's. The same analysis is finding 2 of the repository's `results/mas_continual/interpretation.md`.

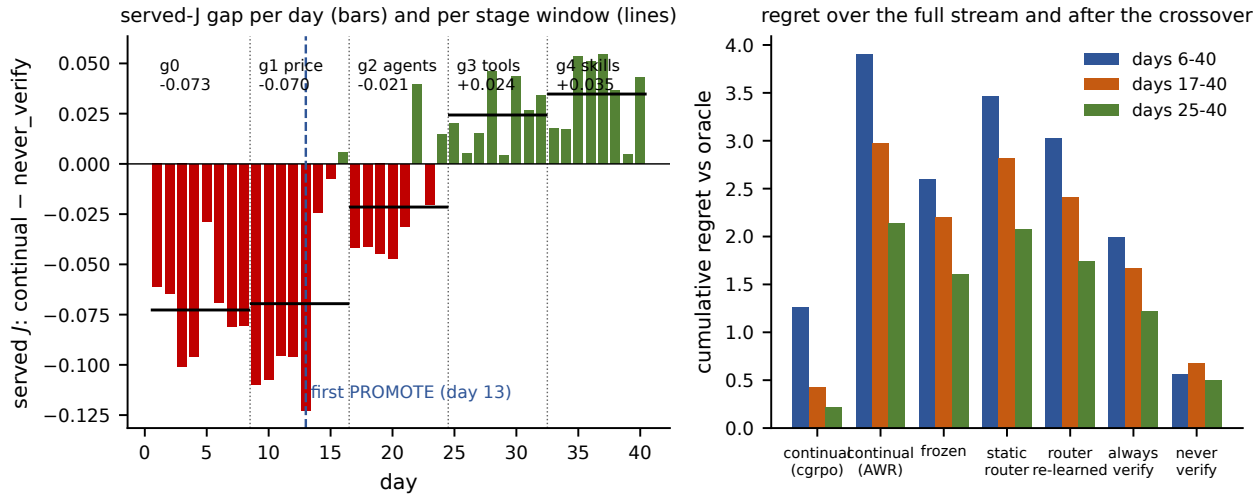


Figure 7: The crossover. Left: served J of the continual arm minus `never_verify`, per day (bars) and averaged per stage window (black lines), with the harness events (dotted) and the first promotion (dashed, day 13). Right: cumulative held-out regret to the oracle over the full stream (days 6–40) and restricted to after the agent-pool event (17–40) and after the tools event (25–40). Both panels from `days.jsonl`.

(3) The gate's price is larger than its benefit on this stream: 32 of 40 candidates reverted, 9 rightly and 23 as false alarms, first promotion on day 13. Figure 8 and Table 12. Every one of the 9 correct REVERTs was a budget violation

(candidate true cost \$0.095–0.101 against $B + \text{tol} = \$0.095$) and no promoted candidate was bad. The DR cost estimate is the criterion that works: RMSE \$0.006 against the simulator's truth, and the set of candidates it stopped is exactly the set of over-budget candidates. The quality criterion produces the false alarms: 18 of the 32 REVERTs failed on quality alone with the candidate acceptable. At $n \approx 500$–1,000 logged episodes the DR quality 90% CI is 0.20 wide and covers the truth 63% of the time, so a candidate whose lower bound must reach the incumbent's point estimate minus 0.05 passes only when it is visibly better. During days 1–12 the served policy was therefore the uniform ϵ-policy while the learner was already good: the frozen arm, which is the learner's day-5 snapshot, scores 0.411 on g1 against the served incumbent's 0.247 on day 9. This is the same sample-size finding as Section 4.3: the cost side is certifiable at about 500 episodes a day, the quality side is not.

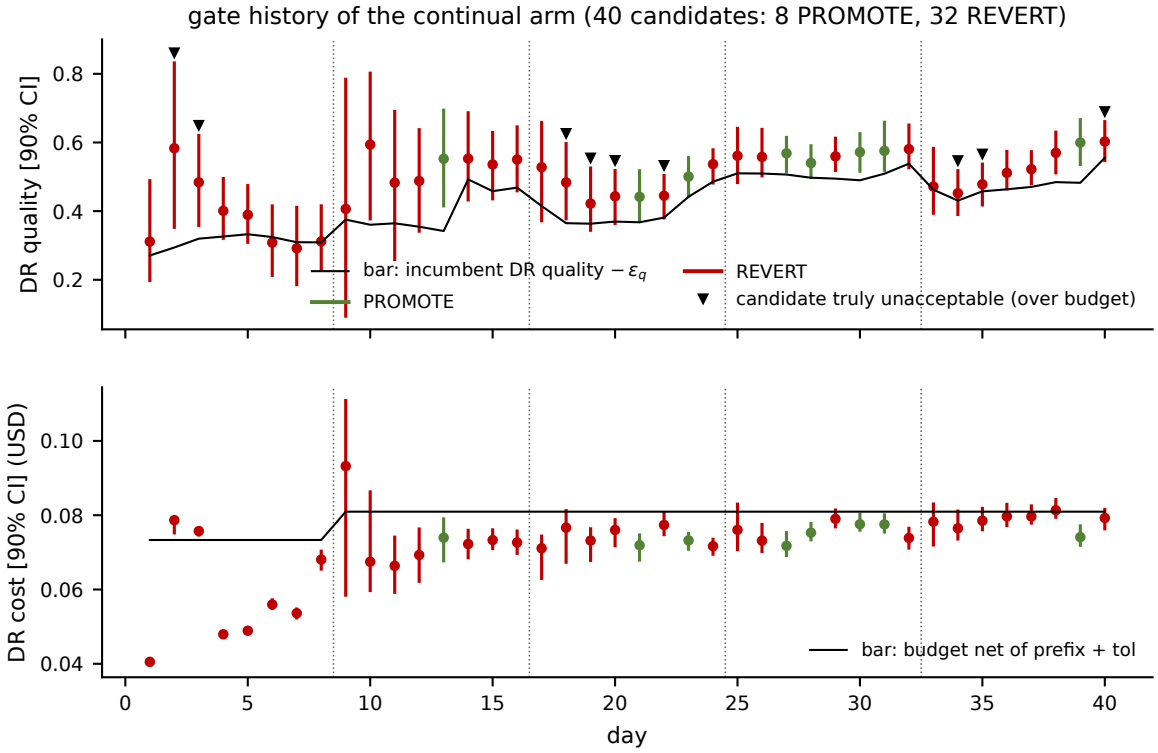


Figure 8: Gate history of the continual arm (from `gates.jsonl`). Top: DR quality with its 90% CI against the bar (incumbent's DR quality minus ϵ_q); bottom: DR cost against the budget net of the prefix plus tolerance. Green is PROMOTE, red REVERT; triangles mark candidates the simulator says were truly over budget. The cost criterion catches every bad candidate; the quality criterion rejects acceptable ones until they are visibly better.

Table 12: Gate history and estimator accuracy against simulator truth (the candidate's true held-out value on the current stage), days after warm-up (35 gates, mean $n = 370$ episodes per gate).

	continual (cgrpo on replay)	continual (AWR+IPS)
candidates: PROMOTE / REVERT	8 / 32	14 / 26
REVERT right (candidate bad) / false alarm	9 / 23	0 / 26
PROMOTE right / missed (candidate bad)	8 / 0	14 / 0
mean per-decision ESS / n	0.54	0.25
mean DR quality CI width / cost CI width	0.200 / \$0.0079	0.653 / \$0.0144
DR quality: bias / RMSE / CI covers truth	−0.005 / 0.067 / 63%	+0.676 / 0.809 / 6%
DR cost: bias / RMSE / CI covers truth	+0.0003 / \$0.0061 / 54%	+0.0000 / \$0.0056 / 69%
per-decision SNIPS quality: bias / RMSE	−0.020 / 0.053	+0.178 / 0.187

(4) The AWR+IPS variant is the cautionary row. It was promoted 14 times and is the worst learned arm (stationary J 0.32–0.36) because the gate's sequential DR over-estimated its quality by +0.68 on average (coverage 6%). AWR's exponentiated inverse-propensity weights move the policy far from the behaviour in one full-batch step; the cumulative ratios (clipped at 50) then carry the estimate with a per-decision ESS of 25% and a bias the clip does not remove. The per-decision estimators' bias on the same candidates is +0.18, better but not a safe bar. The cgrpo candidates, trained on rollouts sampled from the ϵ-mixed served policy, stay close to the behaviour (ESS 54%) and the same estimator is nearly unbiased on them (−0.005). The gate is only as good as the overlap between candidate and behaviour, and a trainer that preserves that overlap is part of the safety story.

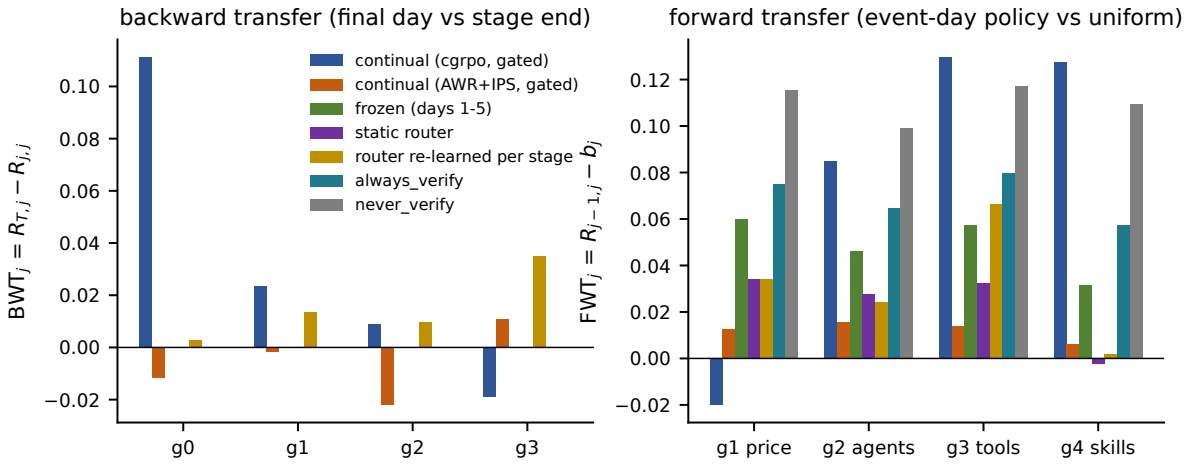


Figure 9: Orchestrator-level backward and forward transfer per stage, Eq. (12), computed from `days.jsonl`. The fixed arms have BWT 0 by construction. The continual arm's large g0 BWT is the day-8 incumbent being the uniform policy (finding 3), not transfer.

(5) Orchestrator-level BWT is bounded and mostly positive. Figure 9. The continual arm's BWT per stage is +0.111 (g0), +0.023 (g1), +0.009 (g2), −0.019 (g3); the mild schedule gives +0.120, +0.026, +0.002, −0.009. Later training did not undo earlier behaviour because the harness only grows: a capability that arrives at stage g has a zero feature slot and a zero logit before it, and the stage-g weights are still what runs where it is not legal. Contrast EvoHarnessBench's −34.7% BWT for agent-pool expansion under a fixed policy: here the pool expansion costs the fixed policies at the event (frozen hit −0.041, static −0.033 at g2) and they never recover (stationary J falls from 0.411 / 0.385 at g1 to 0.350 / 0.316 at g4), while the policy that moves loses nothing on the old stages.

(6) FWT is a property of the policy in service, not of anticipation. New capabilities arrive with zero logits, so the event-day policy is the previous stage's; the continual arm's FWT on g2–g4 (+0.085 / +0.130 / +0.128) exceeds the frozen arm's (+0.046 / +0.057 / +0.032) because it is a better policy, and the static router's FWT falls to −0.002 at g4 because its handbook prices unseen capabilities at a prior. Hits at the events are small for every arm except the price event, where the halved mid price lifts every mid-heavy policy by +0.05 to +0.10.

(7) The SkillOrchestra-style router is the weakest design-time baseline here, and the reason is diagnostic. The competence of an action is the mean outcome of episodes in which it was taken, which under per-step exploration is confounded by everything else in the episode: the handbook over-selects `verify@verify-1` (28% of decisions), an action the real run showed buys nothing. And λ_c is tuned by episode-level SN-IPS of a deterministic router against $\epsilon = 0.1$ traffic, whose ESS is 1–8 of 96, so the choice of λ_c is noise and the re-learned router's λ_c swings between 0 and 20 across stages. Re-learning per stage recovers some of the agent-pool loss (+0.043 at g4) at the price of three exploration days per stage, and still ends 0.07–0.11 below the oracle. This is a statement about applying the competence-minus-priced-cost rule to multi-step orchestration traces with a per-step router, not about SkillOrchestra's own handbook learning from richer per-skill traces on its own benchmarks.

Mild-schedule sensitivity. Figure 10 and Table 11: the ordering is unchanged with smaller gaps (continual 1.10, `never_verify` 0.39, frozen 1.97, static 2.90). In the mild run a learner exceeds the oracle at g1 (gap +0.006), a reminder that the oracle is a reference, not a proven optimum.

Verifier ablation: when does verification help?

Is "`never_verify` beats everything" an artefact of the measured verifier (uninformative: $P(\text{flag} \mid \text{wrong}) = 0.92$ against $P(\text{flag} \mid \text{correct}) = 0.96$) and of the 8-call cap that a verify-and-retry process fills? Four controlled variants of the same 40-day stream, 5 seeds each (seeds 0–4; arms: continual cgrpo, `always_verify`, `never_verify`, per-stage oracle; about 220 s per run): A, the committed configuration (verify-1, cap 8), which also supplies the seed error bars above; B, the informative verify-2 for every arm from day 1 (per-call $P(\text{fail} \mid \text{good rows})$ 0.58 → 0.114, recall unchanged); C, B plus cap 12; D, C plus verifier calls exempt from the cap. Served numbers are means over all 40 days (the fixed modes do not learn); J uses the stage oracle's $\bar{\lambda}_g$.

Table 13: Verifier ablation: `always_verify` minus `never_verify`, served, mean ± SE over 5 seeds.

variant	Δ expected quality	ΔJ	Δ cost / task	cap hits (always)	retries / ep. (always)	verifier calls / ep. (always)
A: verify-1, cap 8 (committed)	-0.033 ± 0.001	-0.036 ± 0.001	+\$0.0036	95%	0.89	2.94
B: verify-2 for every arm	-0.004 ± 0.001	-0.007 ± 0.001	+\$0.0030	91%	0.64	2.90
C: B + cap 12	$+0.029 \pm 0.001$	$+0.011 \pm 0.001$	+\$0.0148	9%	0.76	3.65
D: C + verifier exempt from cap	$+0.028 \pm 0.001$	$+0.010 \pm 0.001$	+\$0.0148	0%	0.77	3.64

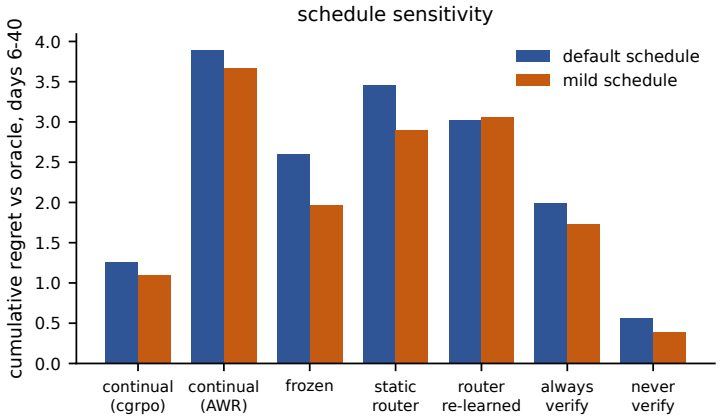


Figure 10: Cumulative regret under the default and the mild schedule. Same ordering, smaller gaps.

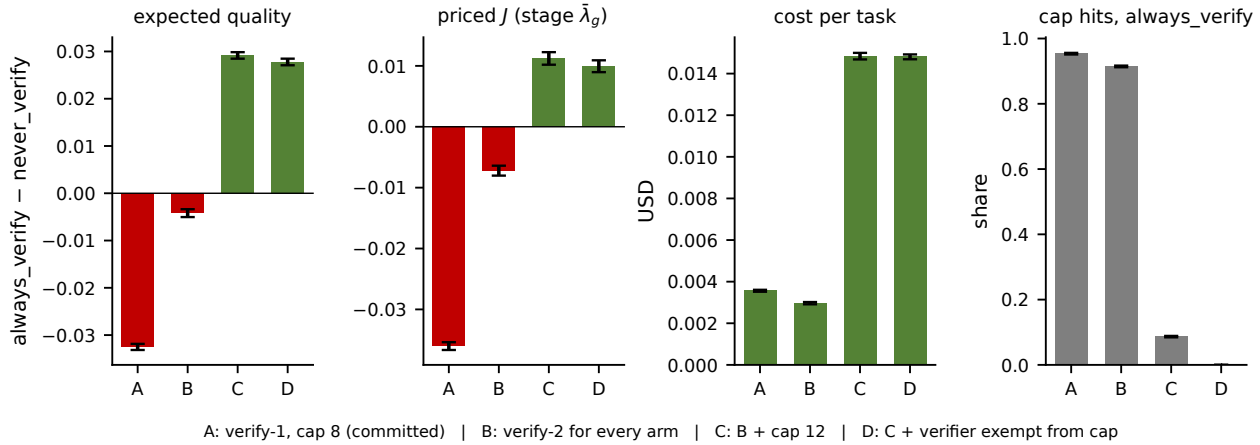


Figure 11: Verifier ablation (from the ablation summary file): `always_verify` minus `never_verify` in expected quality, priced J and cost, and `always_verify`'s cap-hit share, under the four variants; bars are means over 5 seeds with SE. The sign flips only when the verifier is informative *and* the retries do not come out of the plan's call budget.

Table 13 and Figure 11. Under the committed configuration verification *hurts*: `always_verify` is below `never_verify` by 0.033 ± 0.001 in expected quality and 0.036 ± 0.001 in J for +\$0.0036 per task, consistent with the real run's +1 task ($+0.019 \pm 0.067$ SE at $n = 50$). The informative verifier alone (B) closes the gap to -0.004 ± 0.001 but does not reverse it, because 91% of `always_verify` episodes still hit the cap and the plan is truncated. Raising the cap to 12 (C) flips the sign: $+0.029 \pm 0.001$ quality and $+0.011 \pm 0.001$ J for +\$0.0148 per task (+18%), with 3.65 verifier calls and 0.76 retries per episode and 9% cap hits (the stage-4 four-step plans). Exempting verifier calls from the cap (D) changes nothing beyond removing the last cap hits (+0.028 / +0.010): with one retry per sub-query a three-step plan already fits in 12 calls. Stated plainly: verification helps only when the verifier's fail is informative *and* the retry does not come out of the plan's call budget, and then it is worth about +0.03 accuracy for +18% cost, a priced gain of +0.01 J at the stage prices. Under C and D the fixed `always_verify` costs \$0.098–0.105 per task, above the learner's $B + \text{tol} = \$0.095$, so it can exceed the budget-constrained oracle at a stage (g1 in C: 0.443 vs 0.434); the learner's verify share rises from 0.03–0.09 (A) to 0.10–0.13 (B–D) of its decisions and no further, and within budget it still ends the last stage above `never_verify` under every variant (g4: A $+0.031 \pm 0.007$, B $+0.017 \pm 0.011$, C $+0.016 \pm 0.007$, D $+0.013 \pm 0.010$). The whole-stream regret ordering under C/D is `always_verify` 0.30 < `never_verify` 0.60 < continual 1.05–1.20: at a verifier worth using and a cap that does not truncate, the right fixed mode is `always_verify`, it is over budget, and the budgeted learner is third, its cold start still the whole deficit. Two caveats. The informative verifier's episode-level $P(\text{flag} \mid \text{correct})$ in the simulator is 0.71, not the 0.30 its per-call parameter was derived from, because correct episodes still contain empty and wrong steps that are rightly flagged; the per-call parameter is what the ablation sets. And the effect size rests on the assumed link from a step's latent quality to the answer.

The real log gives the per-step number beside the per-episode one (Table 14): 107 verifier fails, 53 followed by a retry of the same sub-query (the rest hit the cap or the one-retry limit). The retry changed the observation kind in 10 of 53 cases (19%), always from empty/error to rows: a flagged empty, error or invalid step became rows in 10 of 20 cases, so the retry does repair malformed or empty steps; but 33 of the 53 retries were of a step that had returned rows, and 26 of those returned rows again. When the retried step was verified again it passed 11 of 47 times (23%) against 42 of 114 (37%)

Table 14: The real log (130 MAS episodes): what a flagged step's retry did. Observation kind before the flag (rows) against after the retry (columns).

before → after	rows	empty	error	invalid	respond	total
rows	26	2	5	0	0	33
empty	5	7	0	1	0	13
error	5	1	0	0	0	6
invalid	0	0	0	0	1	1

for first attempts. Paired with `never_verify` on the same tasks, the 44 `always_verify` episodes that retried flipped 6 wrong→right and 5 right→wrong, with 16 unchanged right and 17 unchanged wrong (25% flipped, symmetric). Per step and per episode say the same thing: the measured verifier's flag is not information about correctness.

What this does not show. Effect sizes are the simulator's: the stage-2+ capabilities are assumed, as is the link from a worker step's latent quality to the answer. One schedule plus a mild one; the all-arms run is one seed, with 5-seed error bars for the continual, `always_verify`, `never_verify` and oracle arms from the ablation of Section 4.7. The per-stage oracle is constrained GRPO on fresh exploration, not a proven optimum. Nothing here calls an API; Phase B (the real-run protocol) is a plan with a cost table (Section 5).

4.8 The same analysis at 150 tasks

After the 50-task results above, the collection was resumed on the same seeded sample to its full 150 tasks (450 episodes, every task on every tier; $59.90 cumulative on the ledger; one interruption at 128 tasks when the API account's credit balance ran out, resumed without loss). Everything in Section 4.2 was re-run on that log with the same code and folds (`results/crmarena_150/`; an intermediate run at 127 tasks is kept in `results/crmarena_127/` and agrees in every ordering).

held-out, 5-fold ($\lambda = 2$ per dollar)	quality	cost/task	J [fold SD]
fixed cheap	0.463	\$0.060	0.344 [0.027]
fixed mid	0.533	\$0.120	0.294 [0.107]
fixed frontier	0.616	\$0.143	0.329 [0.115]
AWR+IPS linear, greedy	0.597	\$0.088	0.421 [0.081]
per-category rule (train-fold argmax)	**0.623**	**\$0.082**	**0.460 [0.094]**
constrained, $B = \$0.10$, greedy of the mix	0.644	\$0.105	0.559 [0.080] at fold $\bar{\lambda}$
per-task oracle	0.711	\$0.057	0.597 [0.040]

Table 15: CRMArena-Pro b2b at 150 tasks: held-out tier policies. The learned per-category policy beats every fixed arm held-out on quality and cost at once; the constrained controller exceeds frontier-tier quality at 27% less cost, at its budget.

Three things changed with three times the data. (1) *The learned policy wins held-out.* At 50 tasks no learned controller clearly beat the best fixed arm; at 150 the per-category rule's J of 0.460 is 0.12 above the best fixed arm (cheap, 0.344), and the constrained controller's 0.644 quality at \$0.105 beats the frontier tier's 0.616 at \$0.143. The margin over the best fixed arm is a little over one fold SD, so this is a consistent ordering rather than a certified gap; the category feature stopped memorising once each category held six or seven tasks instead of two or three. (2) *The gate's answer now depends on the draw, which is the point.* On the seed-0 realised one-arm log the DR estimate for the all-task constrained policy is 0.693 [0.584, 0.757] in quality and \$0.091 [0.078, 0.101] in cost, and the gate PROMOTES; on a second realised draw the cost upper bound is \$0.108 and it reverts. Over 200 redraws the promotion rate is 10% (6% at 50 tasks, 8% at 127). A single promotion at this sample size is a one-in-ten event, so the redraw rate, not the seed-0 decision, is the number to report; the cost interval is still about 0.023 wide against a 0.005 margin, and the thousand-task estimate of Section 4.3 stands. (3) The tier means moved as tasks were added (frontier 0.663 → 0.616, mid 0.608 → 0.533, the oracle 0.769 → 0.711): the 50-task sample was an easy draw, which is the sample variation of Section 4.9 seen from the other side, and the reason the held-out comparison, not the tier table, is the result.

What more tasks would buy, left to the next step. The collection stops at the 150-task seeded sample. Two extensions are costed from the measured rate of about \$0.41 per task across the three tiers and left for the next stage of the project. At *200 tasks* (about \$30 more) each category holds six to nine tasks, which tightens the per-category policy's fold variance, and the gate's cost interval narrows from about 0.036 to about 0.029: the held-out ordering would not change, and the gate would still revert on cost. At *300 tasks* (about \$70 more) the interval reaches about 0.023 and the promotion rate over redraws should move into the ten-to-fifteen percent range; this is also the first scale at which the category-difficulty finding of Section 4.5 could be checked on the single-agent log. Certifying the \$0.005 margin itself remains a thousand-task matter under the $1/\sqrt{n}$ assumption, and the full B2B split (2,140 tasks, roughly \$880) is the ceiling of what this benchmark can offer without B2C.

4.9 Three follow-up experiments: delegation credit, meta-learned initialisation, and a skill handbook

RAO-style delegation credit (negative). We added the node reward of Gandhi et al. [9] to the continual orchestrator's trainer: for a `delegate` decision, the per-decision reward is the episode's priced outcome plus λ_d times the mean success of the sub-call it spawned, with a leave-one-out baseline anchored on the root reward. Over five seeds on the committed simulator and schedule, delegation share rose at every stage (from 0.17/0.44/0.50/0.51/0.55 across the five stages without credit to 0.26/0.47/0.57/0.62/0.62 with λ_d=0.5) and served regret over days 6–40 rose from 0.94 ± 0.27 to 1.74 ± 0.18; stationary priced J fell by 0.02–0.04 at every stage after the baseline. Because the credit mode also changes the baseline, we ran the same mode with λ_d=0: regret 0.99 ± 0.24 and unchanged shares. The credit, not the baseline, buys over-delegation. The learner without credit already delegated to new specialists from the stage they appeared; a bonus outside the baseline turns that into a habit.

Meta-learned controller initialisation (positive, caveated). The learner's deficit against the fixed policy was its cold start. Following the meta-learned initialisation of Tandon et al. [35], we learned the controller's initial weights by a first-order meta-loop over the simulator's stage definitions g_0–g_3 (never g_4, never a held-out set; audited per run), then ran the continual arm from that initialisation. Over five seeds, served regret over days 6–40 fell from 1.06 ± 0.21 to 0.47 ± 0.08 (paired -0.59 ± 0.15) and over days 17–40 from 0.76 ± 0.18 to 0.43 ± 0.07; the learner now has lower regret than `never_verify` (0.74 ± 0.08) on the full stream. The gain comes from what is served before and between promotions, not from earlier promotion (first-promotion day unchanged at 5.2 in this arm set), and there is no gain on g_4, the one stage the meta-initialisation never saw. The caveat is structural: the initialisation was trained on the simulator's own stage definitions; on real traffic it would have to be learned from past harness generations only, and whether the gain survives that is untested.

Skill handbook for the cheap tier (null, with the control that showed it). The cheap tier scores 0.444 on the 50 real tasks because it fails schema reasoning, while the mid and frontier tiers solved many of the same tasks. We distilled per-category procedural notes from those successful SOQL episodes with the mid-tier model as writer (leave-one-task-out, so no task saw notes from its own solution; 49 writer calls, $0.27) and served the cheap tier with the notes in its prompt ($2.39). Quality rose to 0.581 ($+0.137 \pm 0.070$ paired). But the 14 tasks that received no notes improved by the same amount as the 36 that did, so we re-ran the cheap tier with no notes: 0.561, a difference from the skilled arm of $+0.020 \pm 0.067$. The handbook did nothing measurable; the original 0.444 was a low draw. What survives is the size of that draw: the same tier on the same 50 tasks moved by $+0.117 \pm 0.062$ between two runs a day apart, which is as large as the gap between tiers in Section 4.2, and a second, independent reason the gate of Section 4.3 refused to certify a margin at this sample size.

5 Discussion

The signal is the bottleneck. The same pattern appears in every experiment. On CRMArena-Pro the per-task oracle is far above every arm, but the only observable the controller has at step 0 is the category, and with 2–3 tasks per category it memorises (Section 4.2). The orchestration reward model finds that process features carry no out-of-fold signal while the category alone has AUROC 0.82 (Section 4.5); the verifier flags nearly everything; the self-refinement probe barely sees whether the cheap answer is wrong (AUROC 0.65), so the best priced policy is "mostly do not refine" (Section 4.6). The LLM controller's GRPO stage had no advantage signal because its rollout groups were identical, and what it did learn was the price (Section 4.4). In the continual study the gate's quality criterion has too little signal at $n \approx 500$ to tell an acceptable candidate from the incumbent, while the cost criterion, whose signal is a sum of logged dollars, works (Section 4.7). The learners are not the limiting factor; the observables and the sample sizes are. On this benchmark the right orchestration signal is known before the first call, and that argues for investing in observables (a verifier trained on logged (task, answer, score) triples; the mid arm's own answer as a check) before investing in policy classes.

When does a learned orchestrator earn its keep? Only when a harness change moves the right fixed policy. On the stream of Section 4.7, the capability-adding stages are where the continual arm's stationary J exceeds the best fixed mode, and they are also where the fixed modes' gap to the oracle grows ($-0.006 \to -0.038$ for `never_verify`; frozen $0.411 \to 0.350$). A repricing moves the fixed policy's cost but not its ranking; a new specialist or tool moves the ranking, and only a policy that can see the new capability can use it. If the harness is stable and someone knows the right fixed mode, the learner's whole-stream regret (1.26) is its cold start, its recovery days and its gate delays, and the fixed mode (0.56) wins; once the pool has grown past what the fixed mode can use (from day 25: 0.22 against 0.50; g4 margin $+0.031 \pm 0.007$ over 5 seeds), the learner wins while paying an exploration tax the fixed mode does not. And which fixed mode is right depends on the harness: with an informative verifier and a cap the retries do not consume, it is `always_verify`, over budget, and the budgeted learner is third (Section 4.7). The decision to deploy a learned orchestrator is therefore a decision about how often the harness will change in ways that re-rank the fixed policies, which is exactly what the design-time harness-evolution work predicts will be often.

Gate power against day size. With binary outcomes and sequential ratios, the DR quality interval is about ± 0.04 or wider at $n \approx 500$ ($\epsilon_q = 0.05$ was needed because the core default of 0.03 rejects the incumbent against itself at this n), while the cost interval is \$0.008 wide and its criterion is the binding one. Two things follow for a real deployment: certify cost and treat quality as a floor against regression, not as an improvement test; and keep the candidate close to the behaviour (the cgrpo arm's ESS of 54% against AWR's 25%), because the estimator's bias, not its variance, is what promoted the bad arm.

What Phase B would need. The same loop on the real orchestrator: days of 20 b2b tasks, the learner trained with AWR+IPS on the day's log (no replay exists on the real harness) and gated with the per-decision estimators, held-out evaluation by re-serving a fixed 20-task set per stage. Stages that can be induced for real: a repricing (free, costs recomputed from logged tokens); a better verifier (swap its prompt or tier; a mid verifier call is about \$0.002 against \$0.001); a new worker or answer backbone (any priced model); a worker swap by tier; a schema-lookup tool (one describe call) or a local aggregation tool (about \$0); skills as steering text (tokens only). Costs from the measured \$0.084 per episode are in Table 16. A \$0.005 margin on a \$0.09 budget needs about 1,000 logged episodes to certify, so at 20 tasks a day the cost criterion would run on a rolling window and still be underpowered, and the quality criterion at $\epsilon_q = 0.05$ needs on the order of 500 episodes. The honest Phase B at \$50 measures the mechanics (what the learner serves, what the gate does, the frozen policy's hit at each stage) rather than certifying improvements.

Table 16: Phase B cost table (from the measured \$0.084 per episode; \$0.076 under mixed-tier exploration). Per day of 20 tasks, one policy, about \$1.70; the two fixed baselines add \$3.40.

protocol	episodes	estimate
10 days × 20 tasks × 1 served policy	200	\$17
the same with `always_verify` and `never_verify` on every day	600	\$50
20 days × 20 tasks × 1 policy, two induced stages, held-out re-serves (20 tasks per stage)	460	\$39
40 days × 20 tasks × 1 policy, four stages, held-out re-serves	900	\$76
40 days, four stages, with the two fixed baselines	2,500	\$210

6 Limitations and threats to validity

- **Fifty tasks.** The real-harness evidence is one b2b collection of 50 tasks (2–3 per category); every held-out number is a mean over 10 tasks per fold, one task is 0.1 of quality, and the across-fold SDs are as large as the differences between policies. The MAS run has 130 episodes on the same 50 tasks; the SSR run 41 tasks. The negative results for the linear controller, the ORM and the refinement gate are statements about this sample size and these observables, not about the approaches.
- **Seeds.** The simulator pipeline uses one seed per stage, the CRMArena folds one split, and each GPU run ran once; the continual study's all-arms run is one seed per schedule, and only the verifier ablation (continual, the two fixed modes and the oracle; Section 4.7) reports 5-seed error bars. The router and AWR arms carry no seed variance.
- **Simulators calibrated to the project's own logs.** The sim of Section 4.1 is hand-written. The continual simulator's process statistics are measured on 130 episodes, but the link from a step's latent quality to the answer and every capability that arrives after stage 0 are assumed (Appendix B); the transfer effect sizes are the simulator's, not CRMArena's. The calibration check reproduces the fixed modes' cost and call counts but lands accuracy at 0.446/0.459 against 0.480/0.460, because the simulated verifier is uninformative, as the measured one is.
- **Assumed stage parameters.** $P(\text{good} \mid \text{rows}) = 0.60/0.75/0.85$ by tier; coverage-to-answer logit slope 1.5; task-level logit noise sd 0.4; `verify-2` at $P(\text{flag} \mid \text{correct}) = 0.30$ with the same recall; `data-4` at rows 0.85 / good 0.80; `answer-4` mid-grade on 8 categories; `schema_lookup` removes error mass from the next step; `aggregate_calc` adds +0.34 coverage on aggregation categories; the three skills add +0.15 $P(\text{good} \mid \text{rows})$ on their categories or re-plan into 3–4 sub-queries; planner v2 plan length $(0.0, 0.1, 0.4, 0.5)$. Tools and skills are priced as a verifier-sized cheap call (711/62 tokens).
- **The SkillOrchestra baseline is the literal rule, not their system.** Competence minus λ_c times mean logged cost, over task categories as skills, learned from per-step exploration traces of a multi-step orchestrator. Their handbook learning from richer per-skill traces on their own benchmarks is not reproduced.
- **The oracle is a reference.** Constrained GRPO on fresh exploration with more iterations, not a proven optimum; regret can be slightly negative.
- **Gate design.** The sim gate fits its direct model in-sample; the bandit gate cross-fits by task; the continual gate uses the sequential DR with cumulative ratios clipped at 50, whose bias on far-from-behaviour candidates is what finding (3) exposes.
- **Data terms.** CRMArena-Pro is CC BY-NC 4.0 (research use), fetched at setup and not copied; the committed trace is a derived research artefact under the same terms (Appendix E).

7 Related work

Cost-aware routing. FrugalGPT [5] cascades models by a learned score to cut cost; RouteLLM [26] learns a router between a strong and a weak model from preference data; xRouter [29] trains a cost-aware orchestration system with reinforcement learning. These choose a model per query at the first turn; this report prices every decision point inside a frozen harness, learns the price as a dual variable against a budget, and gates deployment off-policy.

Self-evolving agent systems and the deployment seam. A recent position paper from the AReaL group [41] argues that enterprise self-evolving agents are blocked not by RL algorithms but by systems: a step-level data protocol for agent episodes, a data proxy that turns production workloads into governed learning material, and a unified *evolution control plane* that selects, from episode statistics, drift indicators and cost, among memory updates, skill updates, harness edits, weight updates, rollback or no-op, each behind replay-first evaluation, regression tests and canary rollout. That is the architecture this report instantiates in part: the gate of Section 4.3 is a measured, interval-valued form of their replay-first check; priced backward and forward transfer are the drift indicators they call for; and the continual orchestrator of Section 4.7 is a per-request learner inside the harness surface they name, with the harness-edit surface left to the slow loop. Their prototype realises the weight-update path by redirecting deployed agents' inference calls through managed workers, the same seam that Agent Lightning [24] uses to make training harness-agnostic through a proxy that emulates the inference APIs; placing the controller in that proxy is the deployment path we would take, since tier choice, verification and stopping are decisions at the API boundary, while replan and delegation still need a hook inside the orchestrator. Production precedent for the cadence exists: online RL after deployment has shipped with checkpoints refreshed from user interaction every few hours.

Recursive delegation and priced effort. Recursive Agent Optimization [9] trains agents that spawn instances of themselves through a `launch_subagent` call, with a local node reward equal to the node's own success plus λ_d times the *mean* success of its children (averaging rather than counting, which removes the incentive to spawn), a leave-one-out baseline anchored on the root reward, and depth-level inverse-frequency weighting so that deep sub-tasks do not dominate the update. Four points carry over. First, that node reward is the principled process reward for the delegate action of Section 4.5, where our process rewards were definitional; we adopt its form in the continual study. Second, depth weighting is the recursive counterpart of the per-decision estimators used here. Third, delegation in RAO is unpriced: depth and step caps stand in for a budget, so adding $\lambda \cdot \text{cost}$ and $\mu \cdot \text{latency}$ to the node reward (their sequential case runs $18\times$ slower in wall-clock) is the one-line extension that turns the caps into a learned price, and the comparison it sets up is in-weights delegation against controller-side delegation at equal cost. Fourth, RAO's generalization from medium to hard tasks concerns the learned decomposition skill, not drift of the harness around it, which is what the transfer metrics here measure. On the shaping of the price itself, the Composer 2 report describes a nonlinear effort penalty over tokens, tool calls and turns whose slope is steeper on easy tasks than on long-horizon ones; the objective here uses a single linear λ, and a difficulty-conditioned price, λ per task category or a concave cost curve, is a direct extension that the 50-task log is too small to fit.

Contextual bandits in production. The Decision Service [1] and Horizon [10] established the logging discipline (propensities with exploration), the offline evaluation [4, 22] and the promotion workflow that the gate here follows; the estimators are those of Dudík et al. [8], Jiang and Li [14], Swaminathan and Joachims [34], Thomas and Brunskill [37], with cross-fitting after Chernozhukov et al. [7].

Test-time compute and self-refinement. Socratic Self-Refine [32] decomposes an answer into verifiable sub-questions and rewrites where confidence is low; the uncertainty signals (verbalised confidence, sample agreement) follow Bakman et al. [3]. Section 4.6 treats how much of it to buy per request as a priced decision and finds that, on a CRM benchmark, a better tier up front dominates.

Process verification and reward modelling for multi-agent systems. MAS-ProVe [39] studies process-level verification of multi-agent executions; reward modelling for orchestration [38] scores episodes to train orchestrators; MAS-Orchestra [15] asks when a multi-agent system beats a compute-matched single agent. Section 4.5 measures all three on one log and finds the verifier uninformative, the process features without out-of-fold signal, and the single agent ahead.

Harness evolution and continual-learning metrics. The design-time line [2, 6, 12, 21, 43, 44] and EvoHarnessBench [16] evolve or grow the harness; SkillOrchestra [40] learns a static competence router. The BWT/FWT definitions are those of Lopez-Paz and Ranzato [23], applied here to a priced objective at the orchestrator level.

Offline and constrained RL. CQL [19] and IQL [18] are the standard pessimistic offline learners; AWR [27] is the simpler weighted-regression trainer used here with inverse propensities; GRPO [31] supplies the group-relative advantage; the learned price follows RCPO [36] and the PID-Lagrangian [33], with iterate averaging after Nedić and Ozdaglar [25].

Acknowledgements

The authors of CRMArena-Pro released the benchmark, its prompts, evaluator and read-only sandbox that made the real-harness experiments possible.

References

[1] Alekh Agarwal, Sarah Bird, Markus Cozowicz, Luong Hoang, John Langford, Stephen Lee, Jiaji Li, Dan Melamed, Gal Oshri, Oswaldo Ribas, Siddhartha Sen, and Alex Slivkins. Making contextual decisions with low technical debt. *arXiv preprint arXiv:1606.03966*, 2016.

[2] Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*, 2025.

[3] Yavuz Bakman, Duygu Nur Yaldiz, Sungmin Kang, Tuo Zhang, Baturalp Buyukates, Salman Avestimehr, and Sai Praneeth Karimireddy. Reconsidering LLM Uncertainty Estimation Methods in the Wild. *arXiv preprint arXiv:2506.01114*, 2025.

[4] Léon Bottou, Jonas Peters, Joaquin Quiñonero-Candela, Denis X. Charles, D. Max Chickering, Elon Portugaly, Dipankar Ray, Patrice Simard, and Ed Snelson. Counterfactual reasoning and learning systems: The example of computational advertising. *Journal of Machine Learning Research*, 14:3207–3260, 2013. arXiv:1209.2355.

[5] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

[6] Tingyang Chen, Shuo Lu, Kang Zhao, Weicheng Meng, Hanlin Teng, Tianhao Li, Chao Li, Xule Liu, Jian Liang, Zhizhong Zhang, Yuan Xie, Heng Qu, Kun Shao, and Jian Luan. HarnessX: A Composable, Adaptive, and Evolvable Agent Harness Foundry. *arXiv preprint arXiv:2606.14249*, 2026.

[7] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. arXiv:1608.00060.

[8] Miroslav Dudík, John Langford, and Lihong Li. Doubly robust policy evaluation and learning. In *Proceedings of the 28th International Conference on Machine Learning (ICML)*, 2011. arXiv:1103.4601.

[9] Apurva Gandhi, Satyaki Chakraborty, Xiangjun Wang, Aviral Kumar, and Graham Neubig. Recursive Agent Optimization. *arXiv preprint arXiv:2605.06639*, 2026.

[10] Jason Gauci, Edoardo Conti, Yitao Liang, Kittipat Virochsiri, Yuchen He, Zachary Kaden, Vivek Narayanan, Xiaohui Ye, Zhengxing Chen, and Scott Fujimoto. Horizon: An open source applied reinforcement learning platform. *arXiv preprint arXiv:1811.00260*, 2018.

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.09685.

[12] Chengsong Huang, Zifeng Wang, Rujun Han, Jun Yan, Yanfei Chen, Zoey CuiZhu, Ke Jiang, Peng Xia, Han Yu, Yufan Zhuang, Yifei Ming, Jiaqi Pan, Bhavana Dalvi Mishra, Jiaxin Huang, Burak Gokturk, Tomas Pfister, and Chen-Yu Lee. EnvHarness: Awakening Static Worlds for Agent Learning. *arXiv preprint arXiv:2608.19880*, 2026.

[13] Kung-Hsiang Huang, Akshara Prabhakar, Onkar Thorat, Divyansh Agarwal, Prafulla Kumar Choubey, Yixin Mao, Silvio Savarese, Caiming Xiong, and Chien-Sheng Wu. CRMArena-Pro: Holistic assessment of LLM agents across diverse business scenarios and interactions. *arXiv preprint arXiv:2505.18878*, 2025.

[14] Nan Jiang and Lihong Li. Doubly robust off-policy value evaluation for reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016. arXiv:1511.03722.

[15] Zixuan Ke, Yifei Ming, Austin Xu, Ryan Chin, Xuan-Phi Nguyen, Prathyusha Jwalapuram, Jiayu Wang, Semih Yavuz, Caiming Xiong, and Shafiq Joty. MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning Through Holistic Orchestration and Controlled Benchmarks. *arXiv preprint arXiv:2601.14652*, 2026.

[16] Zixuan Ke, Vaidehi Patil, Haizhou Shi, Yang Li, Ye Liu, Sarath Shekkizhar, Anurag Koul, Jiayu Wang, Xuan Phi Nguyen, Semih Yavuz, Mohit Bansal, and Shafiq Joty. EvoHarnessBench: Can Your Agents Keep Pace with an Evolving Harness? *arXiv preprint arXiv:2609.04280*, 2026.

[17] Leslie Kish. *Survey Sampling*. John Wiley & Sons, 1965.

[18] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit Q-learning. *arXiv preprint arXiv:2110.06169*, 2021.

[19] Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative Q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems 33*, 2020. arXiv:2006.04779.

[20] Hoang M. Le, Cameron Voloshin, and Yisong Yue. Batch policy learning under constraints. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. arXiv:1903.08738.

[21] Yoonho Lee, Roshen Nair, Qizheng Zhang, Kangwook Lee, Omar Khattab, and Chelsea Finn. Meta-Harness: End-to-End Optimization of Model Harnesses. *arXiv preprint arXiv:2603.28052*, 2026.

[22] Lihong Li, Wei Chu, John Langford, and Xuanhui Wang. Unbiased offline evaluation of contextual-bandit-based news article recommendation algorithms. In *Proceedings of the 4th ACM International Conference on Web Search and Data Mining (WSDM)*, 2011. arXiv:1003.5956.

[23] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems 30*, 2017. arXiv:1706.08840.

[24] Xufang Luo, Yuge Zhang, Zhiyuan He, Zilong Wang, Siyun Zhao, Dongsheng Li, Luna K. Qiu, and Yuqing Yang. Agent Lightning: Train ANY AI Agents with Reinforcement Learning. *arXiv preprint arXiv:2508.03680*, 2025.

[25] Angelia Nedić and Asuman Ozdaglar. Subgradient methods for saddle-point problems. *Journal of Optimization Theory and Applications*, 142(1):205–228, 2009.

[26] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024.

[27] Xue Bin Peng, Aviral Kumar, Grace Zhang, and Sergey Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. *arXiv preprint arXiv:1910.00177*, 2019.

[28] Doina Precup, Richard S. Sutton, and Satinder Singh. Eligibility traces for off-policy policy evaluation. In *Proceedings of the 17th International Conference on Machine Learning (ICML)*, 2000.

[29] Cheng Qian, Zuxin Liu, Shirley Kokane, Akshara Prabhakar, Jielin Qiu, Haolin Chen, Zhiwei Liu, Heng Ji, Weiran Yao, Shelby Heinecke, Silvio Savarese, Caiming Xiong, and Huan Wang. xRouter: Training Cost-Aware LLMs Orchestration System via Reinforcement Learning. *arXiv preprint arXiv:2510.08439*, 2025.

[30] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

[31] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

[32] Haizhou Shi, Ye Liu, Bo Pang, Zeyu Leo Liu, Hao Wang, Silvio Savarese, Caiming Xiong, Yingbo Zhou, and Semih Yavuz. SSR: Socratic Self-Refine for Large Language Model Reasoning. *arXiv preprint arXiv:2511.10621*, 2025.

[33] Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID lagrangian methods. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. arXiv:2007.03964.

[34] Adith Swaminathan and Thorsten Joachims. The self-normalized estimator for counterfactual learning. In *Advances in Neural Information Processing Systems 28*, 2015.

[35] Arnuv Tandon, Karan Dalal, Xinhao Li, Daniel Koceja, Marcel Rød, Sam Buchanan, Xiaolong Wang, Jure Leskovec, Sanmi Koyejo, Tatsunori Hashimoto, Carlos Guestrin, Jed McCaleb, Yejin Choi, and Yu Sun. End-to-End Test-Time Training for Long Context. *arXiv preprint arXiv:2512.23675*, 2025.

[36] Chen Tessler, Daniel J. Mankowitz, and Shie Mannor. Reward constrained policy optimization. *arXiv preprint arXiv:1805.11074*, 2018.

[37] Philip S. Thomas and Emma Brunskill. Data-efficient off-policy policy evaluation for reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016. arXiv:1604.00923.

[38] King Yeung Tsang, Zihao Zhao, Vishal Venkataramani, Haizhou Shi, Zixuan Ke, Semih Yavuz, Shafiq Joty, and Hao Wang. Reward Modeling for Multi-Agent Orchestration. *arXiv preprint arXiv:2606.13598*, 2026.

[39] Vishal Venkataramani, Haizhou Shi, Zixuan Ke, Austin Xu, Xiaoxiao He, Yingbo Zhou, Semih Yavuz, Hao Wang, and Shafiq Joty. MAS-ProVe: Understanding the Process Verification of Multi-Agent Systems. *arXiv preprint arXiv:2602.03053*, 2026.

[40] Jiayu Wang, Yifei Ming, Zixuan Ke, Shafiq Joty, Aws Albarghouthi, and Frederic Sala. SkillOrchestra: Learning to Route Agents via Skill Transfer. *arXiv preprint arXiv:2602.19672*, 2026.

[41] Ran Yan, Wei Fu, Jiale Li, Shusheng Xu, Zhiyu Mei, Jiaxuan Gao, Jiarui Zhang, Wentai Zhang, Hao Dai, Xujie Shen, Chuyi He, Zhen Pu, Jun Mei, Zhiyao Lin, Haitao Wang, Zhiqiang Ding, Jiawei Zhang, Huaijie Wang, Ruida Xu, Honghua Dong, Youhe Jiang, Yi Wu, Tongkai Yang, and Binhang Yuan. Next-Generation Agentic Reinforcement Learning Systems Enable Self-Evolving Agents. *arXiv preprint arXiv:2607.01120*, 2026.

[42] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.03629.

[43] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin gödel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025.

[44] Yifan Zhang, Yutong Dai, Juntao Tan, Luyu Yang, Rishi Mullur, Thai Hoang, Zhiyuan Hu, James Zhu, Phil Mui, Silvio Savarese, Ran Xu, and Zeyuan Chen. DarwinX: Evolving Agent Harnesses Through Natural Selection. *arXiv preprint arXiv:2608.07545*, 2026.

A Estimator derivations

Unbiasedness of per-decision IPS. For a logged episode drawn from μ, and any step t, $\mathbb{E}_\mu[\rho_{0:t}\, r_t] = \sum_{a_{0:t}} \prod_{k\le t} \mu(a_k \mid s_k) \prod_{k\le t} \frac{\pi(a_k|s_k)}{\mu(a_k|s_k)}\, r_t = \mathbb{E}_\pi[r_t]$, whenever $\mu(a \mid s) > 0$ wherever $\pi(a \mid s) > 0$; the ϵ-mixture of Eq. (4) guarantees the support condition for every served log. Summing over t gives Eq. (8). The clip at $\rho_{\max}$ trades this unbiasedness for variance.

The doubly-robust decomposition. Write $\delta_t = r_t + \hat{V}(s_{t+1}) - \hat{Q}(s_t, a_t)$, the temporal-difference residual of the direct model under the candidate. Eq. (9) is $\hat{V}(s_0) + \sum_t \rho_{0:t}\delta_t$. If $\hat{Q} = Q^\pi$ then $\mathbb{E}_\mu[\rho_{0:t}\delta_t \mid s_t, a_t] = 0$ step by step and the estimate equals $\hat{V}(s_0) = V^\pi(s_0)$ regardless of the ratios; if instead the ratios are exact (no clip) then telescoping the $\hat{V}$ and $\hat{Q}$ terms recovers IPS regardless of $\hat{Q}$. Either model being right is enough [14]. With a clip, the error is $\sum_t (\rho_{0:t} - \tilde{\rho}_{0:t})\delta_t$, which is small when the direct model's residuals are small: the clip biases DR only through the direct model, as stated in Section 2.

Bandit DR with cross-fitting. In Eq. (10), the direct-model term $\sum_a \pi(a \mid x_i)\hat{Q}^{(-i)}(x_i, a)$ is evaluated on a model that did not see task i's reward, so $\mathbb{E}[w_i(r_i - \hat{Q}^{(-i)}(x_i, a_i)) \mid x_i]$ is the reward model's error under π, not a term correlated with r_i by construction. With an in-sample model, $\hat{Q}(x_i, a_i)$ contains r_i itself; with buckets of a few samples the correction term shrinks toward zero and the estimate is pulled toward the direct model. Section 4.3 reports both choices empirically (coverage 99%/92% cross-fitted against 96%/94% in-sample with a -0.006 quality bias).

Why every arm logged makes the estimators exact. With three arms per task each at propensity 1/3, the weights over a task's three episodes are $3\pi(a \mid x)$ and sum to $3\sum_a \pi(a \mid x) = 3$; IPS over the 150 episodes is therefore $\frac{1}{50}\sum_i \sum_a \pi(a \mid x_i) r(x_i, a)$, the exact value, and SNIPS and DR coincide with it. The remaining interval is task-sampling uncertainty.

The ϵ-mixture score. From $\pi_\epsilon = (1-\epsilon)\pi + \epsilon/|\mathcal{A}|$, $\nabla\pi_\epsilon = (1-\epsilon)\nabla\pi = (1-\epsilon)\pi\nabla\log\pi$, so $\nabla\log\pi_\epsilon = \frac{(1-\epsilon)\pi}{\pi_\epsilon}\nabla\log\pi$, Eq. (5). Using the bare score would weight exploration draws as if the policy had chosen them.

Effective sample size. For self-normalised weights, the variance of $\sum_i w_i r_i / \sum_i w_i$ behaves like that of a simple mean over $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ draws [17]; the report flags any estimate with $\mathrm{ESS} < 0.1\, n$.

B Continual simulator: calibrated and assumed parameters

Table 17: Every parameter of the continual simulator with its status (measured from the committed logs, with the count, or assumed).

parameter	value	status	source / note
initial plan length $P(1, 2, 3)$	0.031 / 0.169 / 0.800	measured	130 planner calls (4 / 22 / 104)
re-plan length	1 sub-query	measured	15 re-plans, all emitted 1
data-2 (mid) step kind rows/empty/error/invalid/respond	0.761 / 0.151 / 0.057 / 0.027 / 0.003	measured	331 worker calls
data-1 (cheap) step kind	0.36 / 0.09 / 0.09 / 0.00 / 0.45	measured, thin	11 worker calls
data-3 (frontier) step kind	0.80 / 0.10 / 0.10 / 0.00 / 0.00	measured, thin	10 worker calls
retried step: same kind distribution as a first attempt	rows 0.68 vs 0.76	measured-consistent	53 retries vs 299 first attempts
tokens planner / worker / verifier / answer (in, out)	5,029/279; 8,369/158; 711/62; 12,250/91	measured	ledger means per role
worker input tokens by position	$6833 + 1748k$	measured	positions 0/1/2: 6,833 / 8,571 / 10,330
cheap / frontier tokens per role	planner 6,996/159; 6,902/311; worker 6,647/115; 9,598/196; answer 7,393/245; 15,227/332	measured, thin	5–15 calls each; list prices
verifier $P(\text{fail} \mid \text{empty step})$	0.958	measured	23 of 24
verifier $P(\text{fail} \mid \text{rows step})$	0.581	measured	72 of 124; the same whether the step was good or bad
verifier structural fail on error / invalid steps	1.0 at \$0	measured	12 of 12
answer competence per (agent, category)	logit of MAS per-category accuracy + single-agent (category, tier) offset	measured, shrunk	130 MAS + 150 single-agent episodes; pseudo-count 2.0
max role calls / replans / retries	8 / 1 / 1	measured (configuration)	the real orchestrator's defaults
$P(\text{good} \mid \text{rows})$ per data agent	0.60 / 0.75 / 0.85	assumed	not identifiable from the logs; ordered like the tier accuracies
coverage → answer logit	$1.5 \times (\text{coverage} - 0.571)$	assumed (slope); centre self-calibrated	centre = `never_verify`'s expected coverage
task-level logit noise	sd 0.4	assumed	
worker "respond" on the prefix call	disallowed	assumed	first worker call is always data-2 and always queries
verify-2 (stage 2)	$P(\text{flag} \mid \text{correct})$ 0.30, same recall	assumed	
data-4 (stage 2)	rows 0.85 / good 0.80, cheap price	assumed	
answer-4 (stage 2)	mid-grade on 8 categories, cheap price	assumed	
schema_lookup (stage 3)	next worker step cannot error	assumed	priced as a verifier-sized cheap call
aggregate_calc (stage 3)	+0.34 coverage on aggregation categories	assumed	same price
soql_aggregate, date_window (stage 4)	+0.15 $P(\text{good} \mid \text{rows})$ on their categories	assumed	same price
decompose_fine (stage 4)	re-plan into 3–4 finer sub-queries	assumed	same price
planner v2 plan length $P(1..4)$	(0.0, 0.1, 0.4, 0.5)	assumed	stage 4

Run configuration: 40 days × 96 tasks per arm; $\epsilon = 0.1$; $B = \$0.090$, tol \$0.005, $\epsilon_q = 0.05$; warm-up 5 days; gate window 5; replay window 5; cgrpo 60 iterations, batch 16, group 6; AWR $\lambda = 2$, 200 epochs; 150 held-out tasks per stage; oracle 300 iterations, batch 24, group 8, 200 collected episodes; router refit 3 days at $\epsilon = 0.3$; 200 bootstrap resamples; recovery tolerance 0.02; seed 0; 18 actions; 88 features; runtime 189.1 s (default) and 182.7 s (mild).

C Full per-day tables

Table 18: The continual (cgrpo) arm day by day, generated from `days.jsonl` and `gates.jsonl`. * marks an event day. Served quality is the simulator's expected quality; $\bar{\lambda}$ is the served policy's learned price; ESS is the per-decision effective sample size over logged decisions; "truth" is whether the candidate was acceptable on the simulator.

day	stage	served q	cost	J	$\bar{\lambda}$	gate	DR quality [90% CI]	DR cost hi	ESS	truth
1	g0-baseline	0.405	0.0821	0.302	0.00	REVERT	0.311 [0.198, 0.489]	0.0410	61/199	ok
2	g0-baseline	0.375	0.0776	0.277	0.00	REVERT	0.583 [0.353, 0.832]	0.0790	130/394	BAD
3	g0-baseline	0.361	0.0834	0.256	0.00	REVERT	0.484 [0.358, 0.621]	0.0763	199/606	BAD
4	g0-baseline	0.403	0.0797	0.303	0.00	REVERT	0.401 [0.321, 0.495]	0.0483	298/805	ok
5	g0-baseline	0.408	0.0824	0.305	0.00	REVERT	0.389 [0.309, 0.474]	0.0494	356/991	ok
6	g0-baseline	0.334	0.0813	0.233	0.00	REVERT	0.308 [0.213, 0.415]	0.0572	302/999	ok
7	g0-baseline	0.402	0.0822	0.299	0.00	REVERT	0.292 [0.186, 0.411]	0.0547	292/1013	ok
8	g0-baseline	0.343	0.0814	0.241	0.00	REVERT	0.312 [0.228, 0.415]	0.0703	259/1002	ok
9*	g1-reprice	0.439	0.0802	0.371	0.00	REVERT	0.406 [0.094, 0.784]	0.1109	42/192	ok
10	g1-reprice	0.375	0.0718	0.315	0.00	REVERT	0.594 [0.377, 0.802]	0.0863	81/386	ok
11	g1-reprice	0.379	0.0805	0.311	0.00	REVERT	0.483 [0.259, 0.690]	0.0741	116/587	ok
12	g1-reprice	0.385	0.0755	0.321	0.00	REVERT	0.488 [0.342, 0.637]	0.0763	158/788	ok
13	g1-reprice	0.381	0.0772	0.316	0.00	PROMOTE	0.552 [0.415, 0.694]	0.0790	190/995	ok
14	g1-reprice	0.482	0.0898	0.407	0.81	REVERT	0.553 [0.433, 0.687]	0.0759	254/1070	ok
15	g1-reprice	0.491	0.0874	0.417	0.81	REVERT	0.536 [0.436, 0.629]	0.0761	347/1143	ok
16	g1-reprice	0.479	0.0864	0.406	0.81	REVERT	0.550 [0.459, 0.645]	0.0757	473/1200	ok
17*	g2-agents	0.472	0.0645	0.419	0.81	REVERT	0.528 [0.372, 0.658]	0.0744	143/283	ok
18	g2-agents	0.420	0.0756	0.359	0.81	REVERT	0.484 [0.378, 0.597]	0.0812	199/561	BAD
19	g2-agents	0.449	0.0700	0.392	0.81	REVERT	0.422 [0.344, 0.525]	0.0763	236/829	BAD
20	g2-agents	0.458	0.0699	0.401	0.81	REVERT	0.443 [0.364, 0.518]	0.0788	307/1101	BAD
21	g2-agents	0.470	0.0768	0.408	0.81	PROMOTE	0.442 [0.370, 0.518]	0.0747	408/1395	ok
22	g2-agents	0.531	0.0870	0.461	0.75	REVERT	0.445 [0.380, 0.504]	0.0805	509/1461	BAD
23	g2-agents	0.531	0.0857	0.461	0.75	PROMOTE	0.501 [0.443, 0.556]	0.0751	570/1507	ok
24	g2-agents	0.552	0.0869	0.482	1.03	REVERT	0.537 [0.482, 0.579]	0.0735	900/1593	ok
25*	g3-tools	0.519	0.0883	0.461	1.03	REVERT	0.561 [0.483, 0.641]	0.0830	339/345	ok
26	g3-tools	0.524	0.0875	0.466	1.03	REVERT	0.558 [0.503, 0.638]	0.0775	665/680	ok
27	g3-tools	0.559	0.0914	0.499	1.03	PROMOTE	0.568 [0.515, 0.615]	0.0753	997/1022	ok
28	g3-tools	0.511	0.0788	0.459	0.77	PROMOTE	0.541 [0.499, 0.590]	0.0778	1275/1371	ok
29	g3-tools	0.548	0.0971	0.484	0.66	REVERT	0.560 [0.519, 0.612]	0.0814	1410/1736	ok
30	g3-tools	0.535	0.0910	0.475	0.66	PROMOTE	0.572 [0.516, 0.626]	0.0802	1363/1747	ok
31	g3-tools	0.543	0.0832	0.488	0.53	PROMOTE	0.576 [0.519, 0.659]	0.0801	1368/1774	ok
32	g3-tools	0.561	0.0872	0.503	0.41	REVERT	0.581 [0.527, 0.651]	0.0765	1459/1809	ok
33*	g4-skills	0.518	0.0941	0.449	0.41	REVERT	0.472 [0.394, 0.583]	0.0830	412/432	ok
34	g4-skills	0.470	0.0874	0.407	0.41	REVERT	0.452 [0.390, 0.518]	0.0811	810/848	BAD
35	g4-skills	0.585	0.1041	0.509	0.41	REVERT	0.478 [0.418, 0.537]	0.0818	1165/1277	BAD
36	g4-skills	0.542	0.0986	0.471	0.41	REVERT	0.512 [0.465, 0.574]	0.0829	1540/1720	ok
37	g4-skills	0.539	0.1012	0.465	0.41	REVERT	0.522 [0.480, 0.573]	0.0825	1959/2164	ok
38	g4-skills	0.506	0.0900	0.441	0.41	REVERT	0.570 [0.512, 0.630]	0.0842	1824/2145	ok
39	g4-skills	0.505	0.0926	0.437	0.41	PROMOTE	0.600 [0.536, 0.666]	0.0771	1765/2142	ok
40	g4-skills	0.555	0.0881	0.491	0.88	REVERT	0.603 [0.547, 0.661]	0.0816	1789/2103	BAD

Table 19: Stationary held-out J (mean over the last four days of each stage), default schedule.

arm	g0	g1	g2	g3	g4
oracle	0.345	0.460	0.445	0.458	0.466
continual (cgrpo on replay, gated)	0.283	0.427	0.429	0.453	0.448
continual (AWR+IPS, gated)	0.294	0.363	0.342	0.330	0.321
frozen (trained days 1–5)	–	0.411	0.370	0.374	0.350
static router	–	0.385	0.352	0.349	0.316
router re-learned per stage	–	0.389	0.369	0.346	0.363
`always_verify`	0.329	0.426	0.389	0.397	0.376
`never_verify`	0.368	0.466	0.423	0.434	0.428

Table 20: Candidates the gate stopped that were in fact over budget (continual cgrpo arm; the AWR arm had none).

day	stage	candidate true quality / cost	incumbent true quality / cost	over budget	gate said
2	g0	0.482 / \$0.1009	0.382 / \$0.0794	yes	REVERT
3	g0	0.470 / \$0.0981	0.382 / \$0.0794	yes	REVERT
18	g2	0.515 / \$0.0974	0.467 / \$0.0721	yes	REVERT
19	g2	0.508 / \$0.0953	0.467 / \$0.0721	yes	REVERT
20	g2	0.528 / \$0.0996	0.467 / \$0.0721	yes	REVERT
22	g2	0.527 / \$0.0962	0.512 / \$0.0926	yes	REVERT
34	g4	0.521 / \$0.0966	0.513 / \$0.0914	yes	REVERT
35	g4	0.522 / \$0.0951	0.513 / \$0.0914	yes	REVERT
40	g4	0.523 / \$0.0954	0.517 / \$0.0904	yes	REVERT

Table 21: SkillOrchestra-style handbooks: when fitted, on how many episodes, the chosen λ_c and the SN-IPS estimate that chose it (its ESS of 96 shows why the choice is noise).

handbook	fitted on day	episodes (learn / tune)	λ_c	SN-IPS quality / cost at λ_c (ESS)
static	5	384 / 96	0.0	1.000 / \$0.0787 (4/96)
relearn g1	11	192 / 96	0.0	0.597 / \$0.0911 (7/96)
relearn g2	19	192 / 96	20.0	1.000 / \$0.0278 (1/96)
relearn g3	27	192 / 96	0.0	0.705 / \$0.0407 (3/96)
relearn g4	35	192 / 96	0.0	0.750 / \$0.0690 (8/96)

D Reproduction

All CPU pipelines are seeded end to end and need only `numpy`; the CRMArena-Pro offline script is byte-for-byte deterministic on the committed log. From the repository root:

```
pip install -r requirements.txt
bash scripts/run_v0.sh              # sim: collect -> train -> gate -> drift -> retrain -> gate (~2 min)
python scripts/crmarena_offline.py  # CRMArena-Pro Part A on the committed log, no API calls (~15 s)
python -m pace.experiments.mas.run analyze --log results/mas/mas_b2b.jsonl \
    --single-agent-log traces/crmarena_b2b_counterfactual.jsonl --out results/mas/RESULTS.md
python -m pace.experiments.ssr.run analyze --log traces/ssr/ssr_b2b_tree.jsonl \
    --out results/ssr/RESULTS.md --csv results/ssr/frontier.csv
python -m pace.experiments.mas_continual.run_continual --out results/mas_continual
python -m pace.experiments.mas_continual.run_continual --schedule mild \
    --out results/mas_continual/mild
python -m pace.experiments.mas_continual.analyze --run results/mas_continual \
    --mild results/mas_continual/mild --out results/mas_continual/RESULTS.md
python -m pytest tests -q
```

The GPU runs (`pace/llm/`) each ran once in one A100-40GB container; their training logs, run metadata, SFT reports and evaluation tables are committed under `results/llm/`. The real-harness collections (`pace.collect --harness crmarena --mode counterfactual`, the MAS and SSR `run` commands) require the benchmark checkout, its sandbox credentials and a model API key, and are capped by the spend ledger. The figures in this report are regenerated by `python paper/make_figures.py` from the committed result files, and the report by `make` in `paper/`.

E Data license note

CRMArena-Pro is released under CC BY-NC 4.0 for research use. It is fetched at setup and never copied into the repository. The committed trace `traces/crmarena_b2b_counterfactual.jsonl` is a derived research artefact (per-episode features, actions, costs, the agent's answers and queries, and the benchmark's gold answers for 50 tasks) under the same non-commercial terms; the matching ledger snapshot contains token counts and prices and no benchmark data. The MAS and SSR logs are derived artefacts of the same kind.